



Second lecture: Polysomnography Demonstration/ MSLT

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SCHOOL OF MEDICINE

Sleep is a natural restorative physiological process that is characterized by perceptual disengagement and must be rapidly reversible



“Dream to Remember, Dream to Forget”

Functionsof sleep

- Higher-order cognitive function involving mechanisms of learning and memory and synaptic plasticity



- Restorative function including maintenance of brain energy metabolism, macromolecular biosynthesis, and removal of metabolic waste



**University of Puerto Rico
1980-1984**

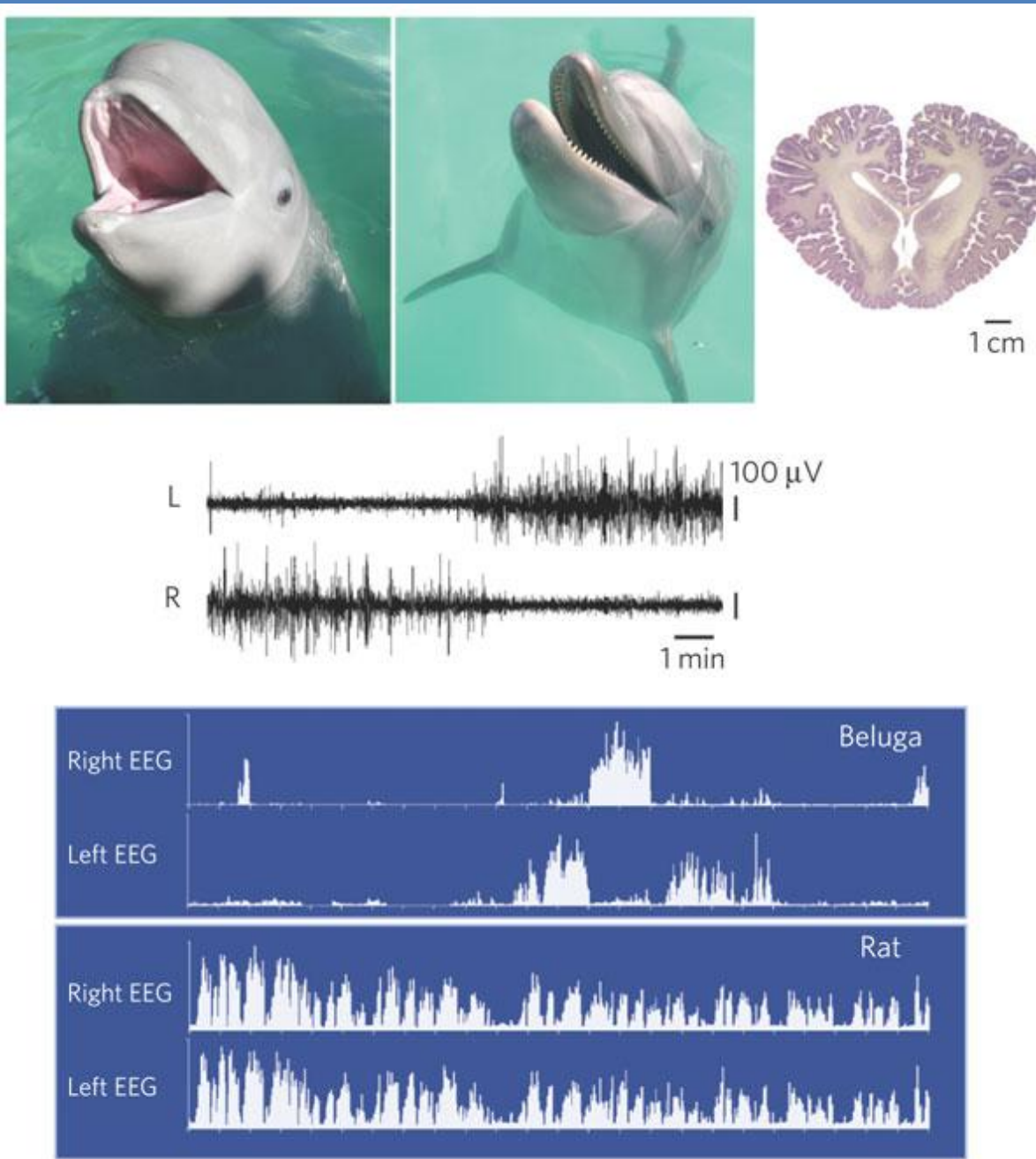
Advocacy and Public Policy

I volunteer with 4 non-profit organizations:

- Chair, American Academy of Sleep Medicine's Political Action Committee:
<https://aasm.org/advocacy/pac/>
- Vice Chair, National Sleep Foundation <https://www.thensf.org/>
- Past President, California Sleep Society <https://californiasleepsociety.org/>
- BOD, Start School Later <https://www.startschoollater.net/>

Dream Characteristics

- Intense emotions
- Hard to remember afterwards
- Illogical/ impossible situation not questioned
- Odd/old memories incorporated into content
- Fascinating to the dreamer



Clues to the function of mamalian
sleep Siegel Nature 05

Early Dreams and Society

- How would a prehistoric society explain why they could see their departed loved ones in a dream?
- The need to explain dreams may be the origin of early religions
- A great leader is said to have a “vision”. That vision will unite the people
- Inspiration of dreams may have led to the establishment of great societies

Why would *Homo naledi* bury their dead?

The ancestor we never knew

The discovery of a human ancestor, *Homo naledi*, in an ancient burial chamber in South Africa has been described by scientists as the most significant find in the study of human evolution for four decades.

Timeline

Where *Homo naledi* could be dated

Today

N. memberensis
A. africanus
A. erectus

One million years ago

Australopithecus boisei
A. robustus
A. habilis
A. rudolfensis

Two million

A. senilis
Homo naledi

Three million

A. africanus
A. garhi
A. sp. "Orrorin"
Ramapithecus platyotus
A. afarensis
Explosion leader Professor Lee Berger

Four million

A. anamensis

Homo naledi

Using bone scans, clay and liberal amounts of beer for an artist has recreated the face of *Homo naledi*.

Skull
Was around half the size of ours at 560 cubic centimetres

Human skull

"Rising Star Hominid"
Homo naledi
Date Unknown
Adult male
48-100cm, 45-50kg

Turkana Boy
Homo erectus
1.6m years ago
Adolescent male
SPH, 90-120kg

"Lucy"
Australopithecus afarensis
3.2m years ago
Adult female
SPH 60, 27-29kg

Carved fingers
An ape-like adaptation, useful in grasping them

Humanlike feet
Designed for standing upright and walking long distances

The caves

The Dinaledi chamber containing the bones translates, in local dialect, to "Chamber of Stars".

Rising star cave

Entrance

Sapien's cave! So named because it is so tight you have to put one arm above your head!

AFRICA

ETHIOPIA
Afar Depression
Discovery of LUCY reveals (50'000)

KENYA
Lake Turkana
Discoveries of Turkana Boy (160k)

MALAWI
Johannesburg

SOUTH AFRICA

Rising star cave

Below: an artist's impression of an *Homo naledi* group dispersing one of their own in the cave

Dreams and Religion

- Most if not all religions discuss dreams
- Dreams are a source of Divine Revelations
- In the book of Genesis, God must rest and Adam is put to sleep to make Eve



**Sumerians recorded dreams over
5,000 years ago on clay tablets**



Ancient Egypt

Dreams viewed as a guiding force and as a journey to the underworld

Special dream beds inside the temples



Hinduism

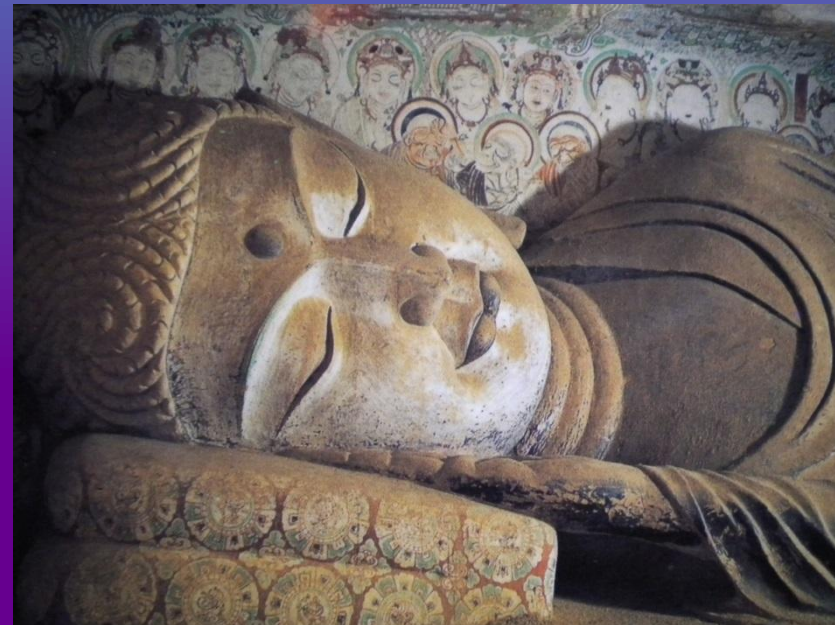


Dreams are real and caused by
the Supreme Brahman



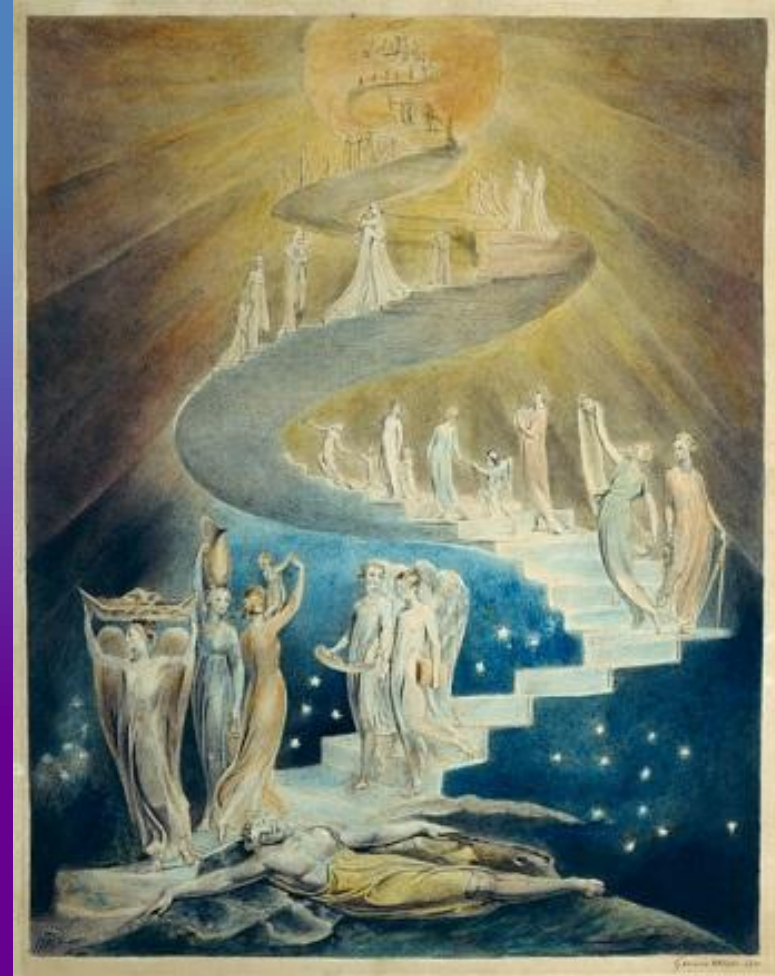


Queen Maya
dreams her son
will be the
Enlighten One



Old Testament

Jacob's dream of the Divine ladder at Bethel (Gen 28:11-19)



New Testament



And Joseph her husband, being a righteous man and not wanting to disgrace her, planned to send her away secretly. But when he had considered this, behold, an angel of the Lord appeared to him in a dream, saying, "Joseph, son of David, do not be afraid to take Mary as your wife; for the Child who has been conceived in her is of the Holy Spirit. Matthew 1:20

The Night Journey and Islam



The Prophet Mohammed travels to Jerusalem and Heaven

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California Child Who Died Was Allegedly Subject to Exorcism

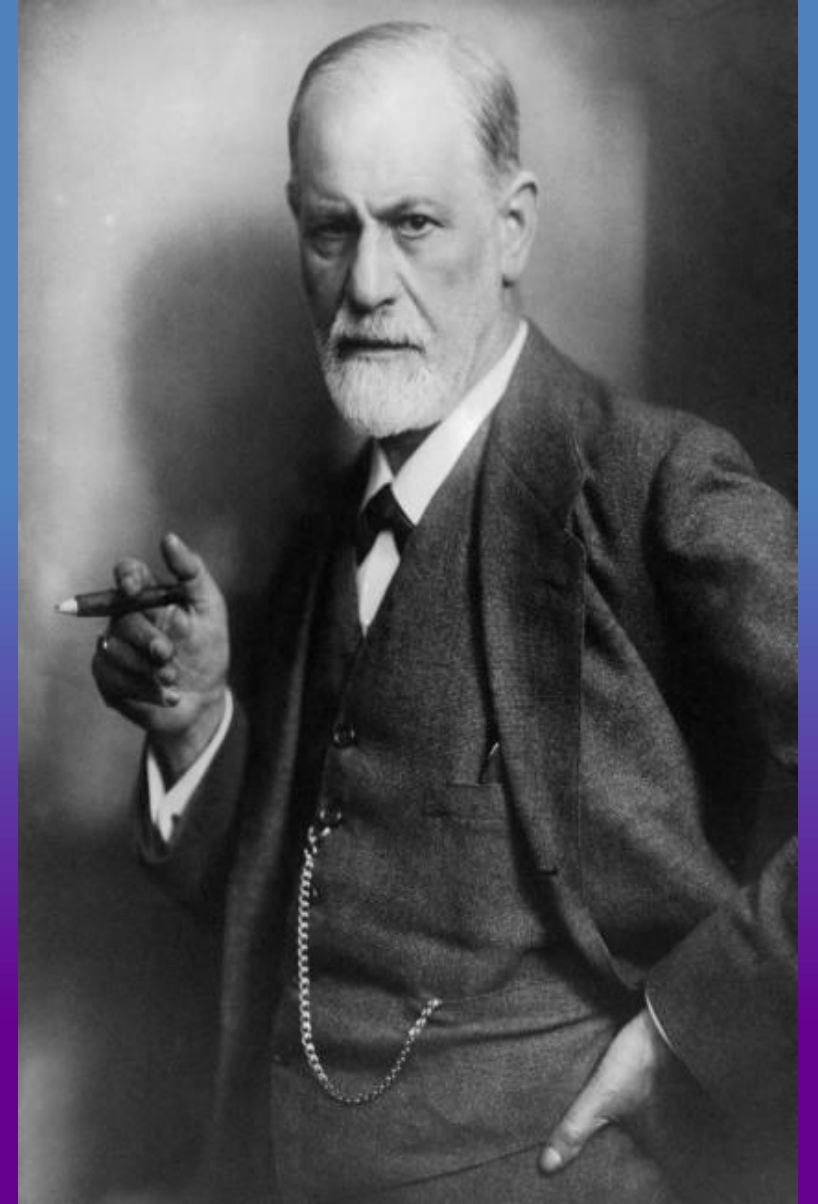
A 3-year-old girl whose death last fall has been ruled a homicide was the subject of an alleged exorcism last year at a Northern California church.

By [Associated Press](#) | May 10, 2022, at 7:38 p.m.

San Jose, California 2022

Royal Road To The Unconscious

- *Interpretation of Dreams* Freud 1899
- The content of dreams is driven by unconscious wish fulfillment. Sexual focus probably reflected the cultural repression. His focused evolved
- 25 years afterwards the human EEG recorded by Hans Berger in 1924



THE PHENOMENA OF SLEEP.



THE cry for sleep is ever greater than the cry for bread. Existence depends on both; but we eat to sleep, while we sleep to live. Sleep is of far greater importance than food for the preservation of life.

Sleep is the chiefest thing in physic. It has neither substitute nor rival. Take away sleep, hope even is gone, nothing is left.

*The
chiefest
thing*

The Witchery of Sleep by Willard Moyer

“Good sleep isn’t just a medical issue. It’s also an ethical one”

AMA Journal of Ethics®

Illuminating the Art of Medicine



Search

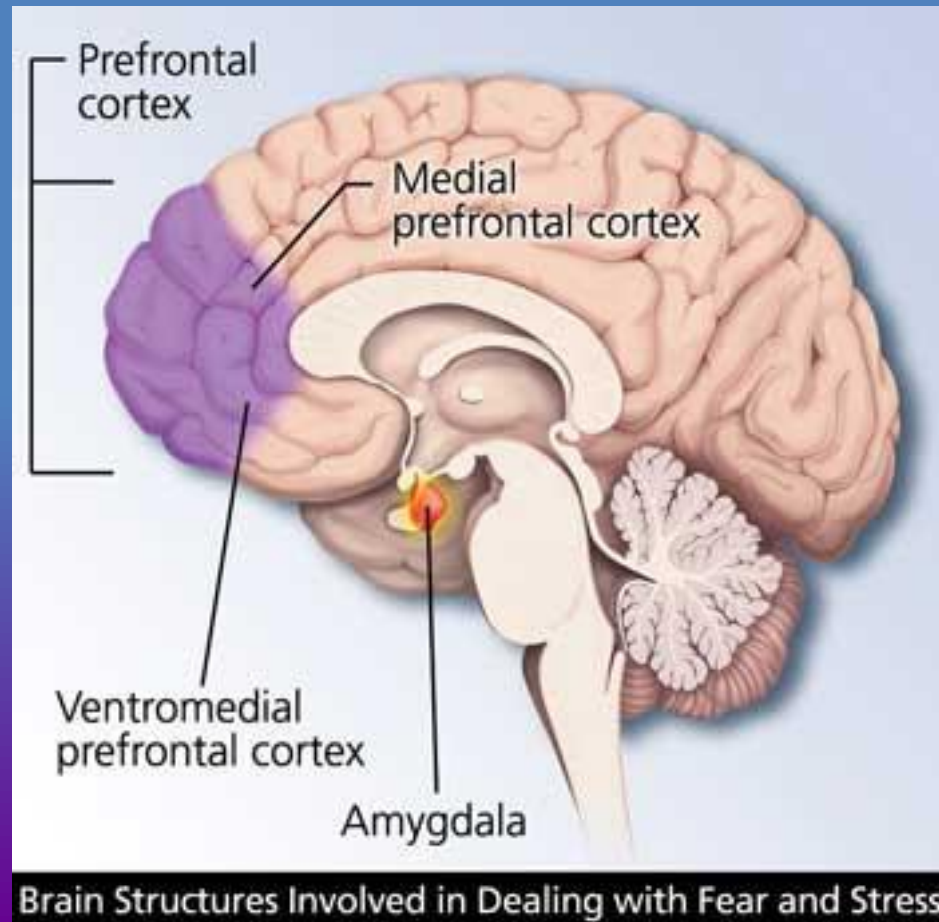
OCT 2024

Sleep Stewardship

Sleep hygiene, causes of sleep disruption, and sleep disorder interventions are recent, important topics of health and cultural awareness. The better one’s sleep quality over their lifespan, the better their health status and health outcomes, so who has reliable access to quality sleep is not just a clinical matter, but one for ethics and justice. This theme issue’s focus is broader than poor sleep pathologies and investigates how sleep is, perhaps, best conceived as a communal, natural resource. We all need clean air and water, shelter, nutritionally dense food, and sleep. Of course, we sleep as individuals, but our common human interest in quality sleep of sufficient duration generates collective obligations to respond equitably to chronic health conditions that exacerbate poor sleep patterns; to support conditions for feeling safe, peaceful, and calm enough to rest; and to mitigate noise and light pollution that compromise our and our neighbors’ sleep environments.

Volume 26, Number 10: E745-821

Why Sleep?



The study of sleep is ultimately a study of consciousness

- In 1875 became the first person to measure the electroencephalogram at the age of 32
- He measured “feeble” electrical activity of exposed brains of cats, monkeys, and rabbits
- He did not have any amplifiers, he watched for changes in mercury flow as he attached an electrode to the brain and another to the skull
- Prior to this electro physiology was mostly limited to work showing the contractions of frog muscles.



**Richard
Canton**
1842-1926

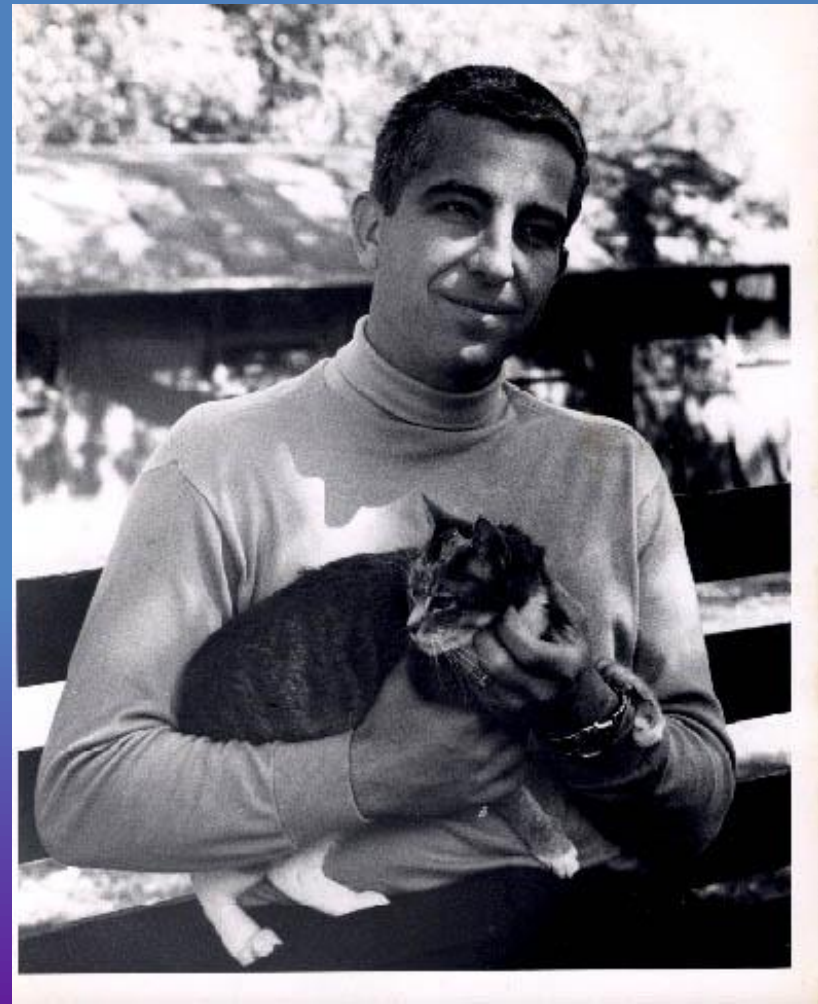
- In 1924-1927 was the first person to measure the EEG of humans and discovered alpha rhythm. He faced skepticism in his native Austria/Germany.
- He believed in telepathy and thought he could transmit his thoughts
- It was not until years later that his work was replicated that he received recognition.
- He was passed over for the Noble prize. History is not clear but there are accounts of him having been stopped from gaining recognition for his work by the Nazi's. Other accounts are that he was a Nazi sympathizer.
- He hung himself in 1941



Hans Berger
1873-1941

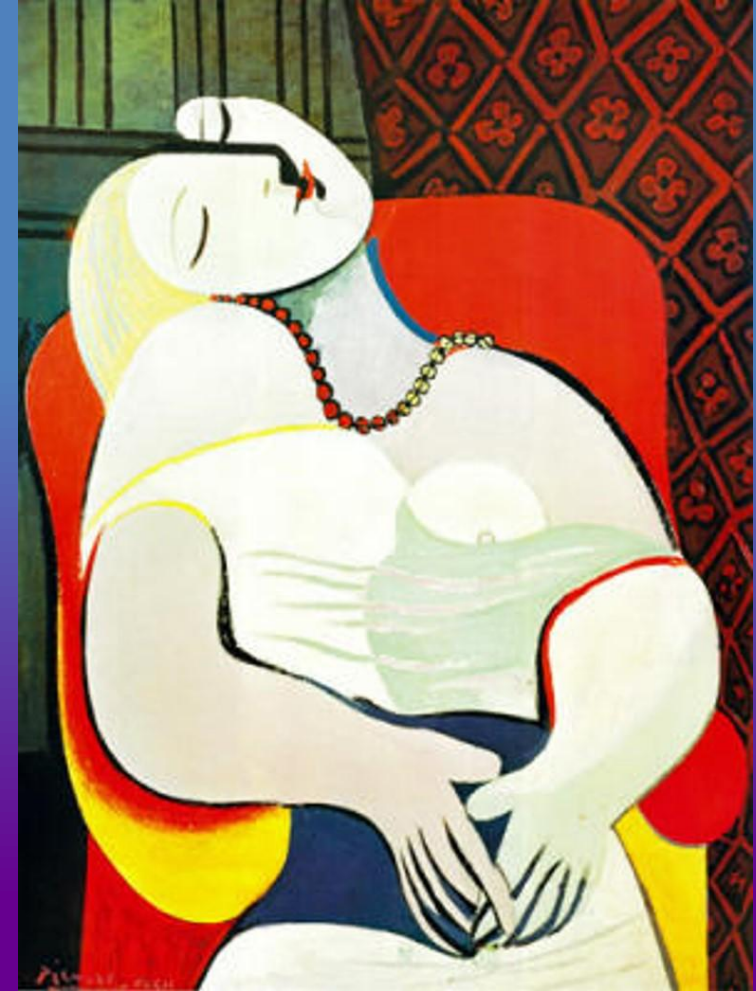
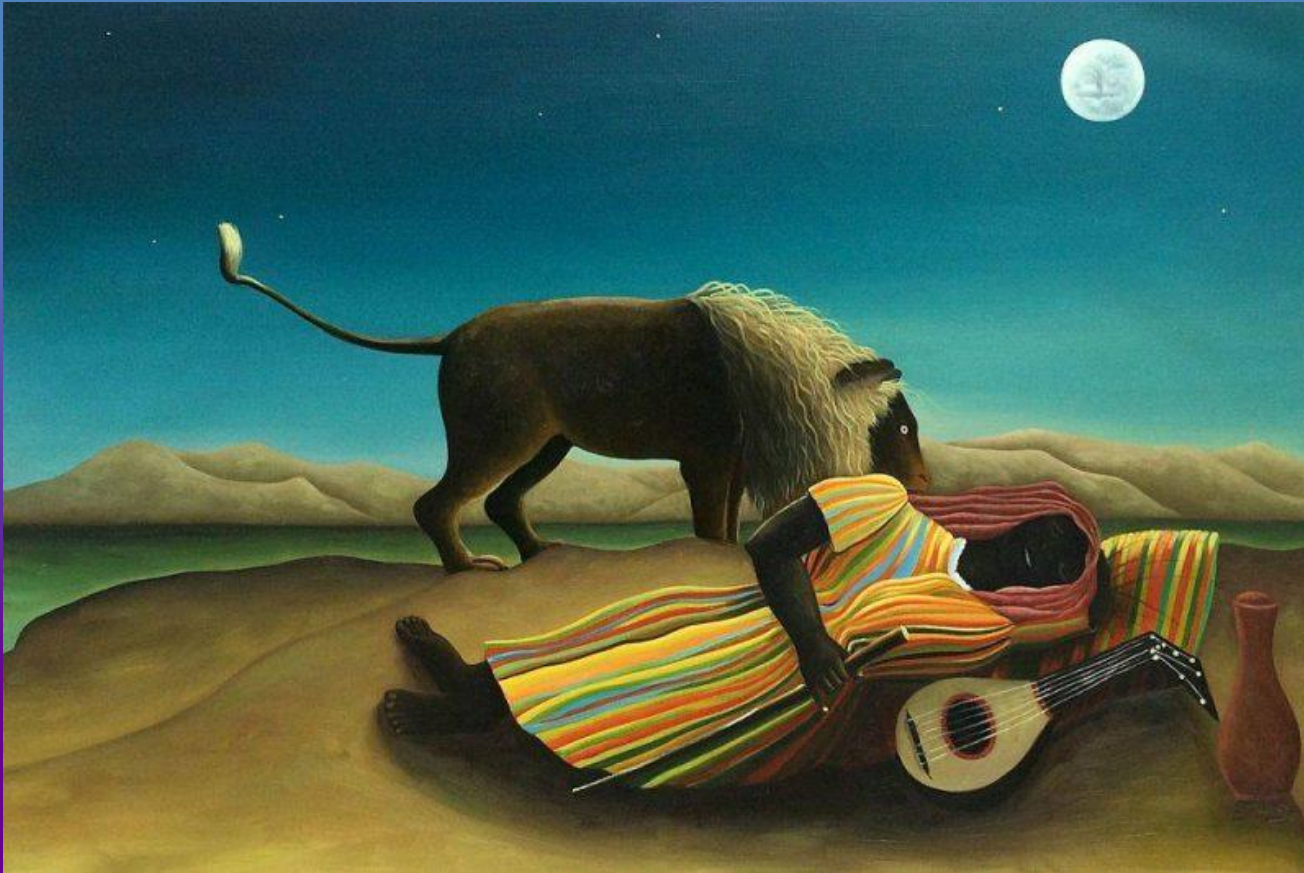


- 1937 Performs the first all night sleep EEG recordings
- **Describes the K-complex**
- Graduated Yale Law in 1912, joins the Army for WW1, he invents ballistics instruments. He becomes rich via insider trading in the 1920's. As a tycoon, anticipates the Stock Market Crash.
- Sets up secret research facilities for the top scientists in his home in NY. During the week he works in Wall Street and on weekends did science.
- Then in WW2 helps invent radar, and helps win the war.



**Kleitman, Aserinsky, and Dement and colleagues
discovered REM link to dreams in 1953**

We sleep to help the brain learn to adapt to a changing environment



**Do you really want to sleep
like a baby?**





Do you remember being tucked in or tucking somebody else in?



You should not wake up feeling tired

The first thing that happens when you don't sleep well is you become irritable and inattentive



Remember being 7 or 8 years old and when you woke up realized you did not have to go to school?

Do you know anybody that wakes up irritable or is inattentive?



**You fall asleep in class because you are
bored: FALSE**

Monotony does not cause sleepiness



Monotony does unmask sleepiness



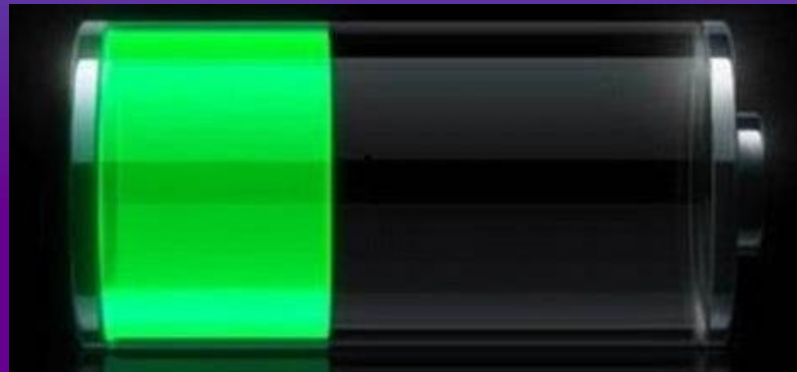
The need for sleep is biological,

The way you sleep is learned

What wakes you up, *may not be what keeps you awake*



How do you feel when you are away and your cell phone is only half charged?



The brain functions in 3 states

	Awake	NREM	REM
ACH	++	-	++
NA	++	+	--
SE	++	+	--

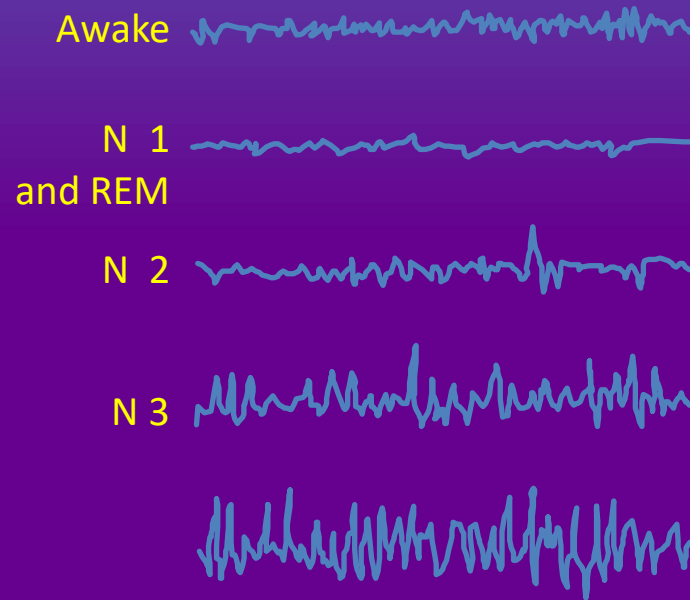


ACH acetylcholine, NA noradrenaline, SE serotonin

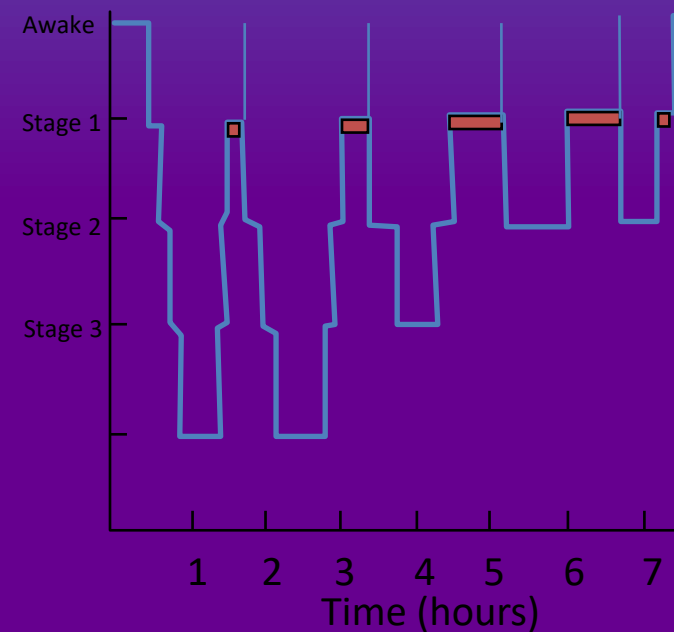
Sleeping through the night is a myth

None of us sleep through the entire night

EEG Recordings

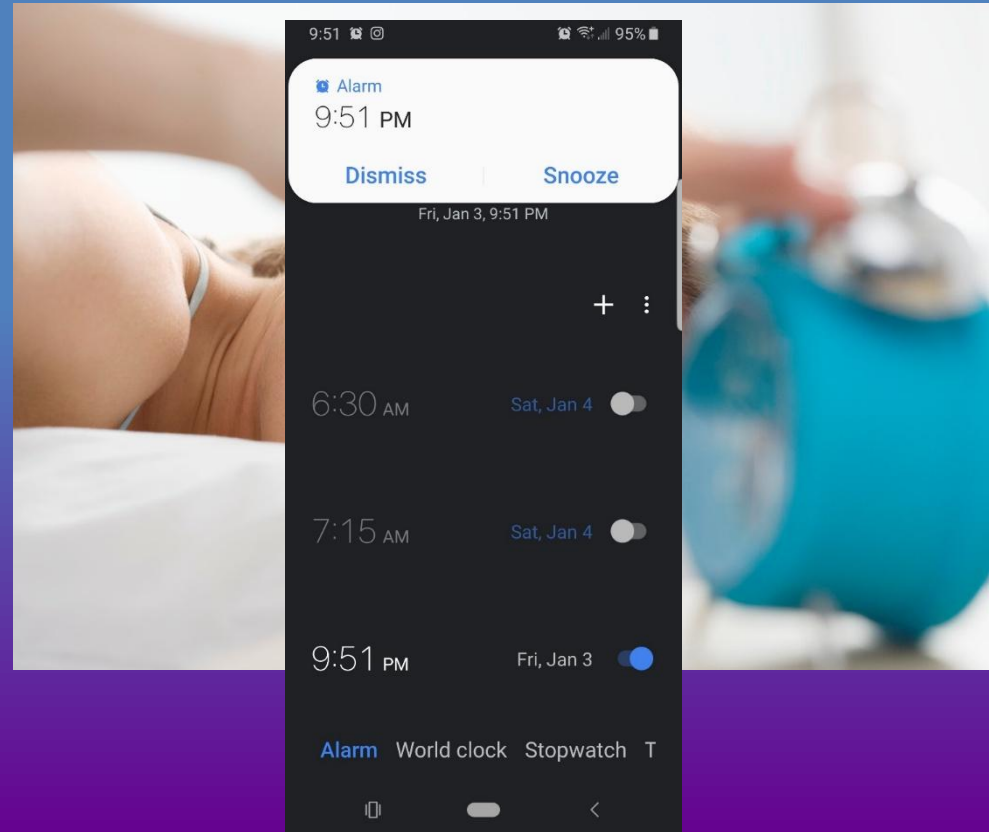


Typical Nighttime Sleep Pattern in Young Adult



When we wake up too early we are not merely sleep depriving ourselves we are specifically dream depriving ourselves

If life is a journey,



the snooze button is a pothole

What Drives Us Toward Sleep and What Keeps Us Asleep?



Sleep Timing

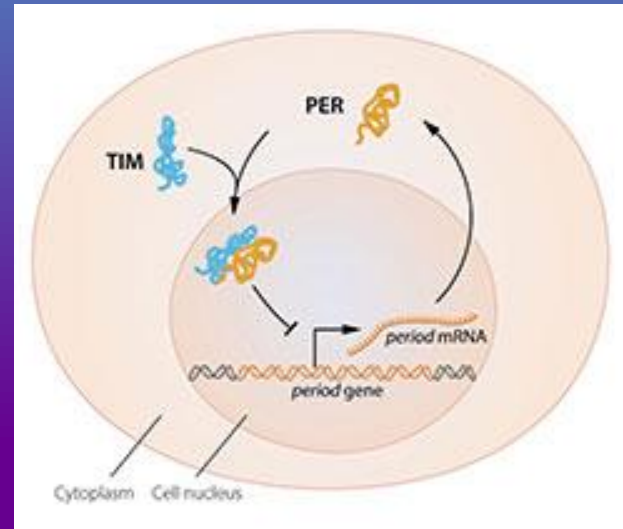
- Sleep timing is influenced by homeostatic and circadian factors
- The less we sleep the more sleep we need and vice versa
- Twice a day our alertness level peaks
- Twice a day our sleepiness peaks

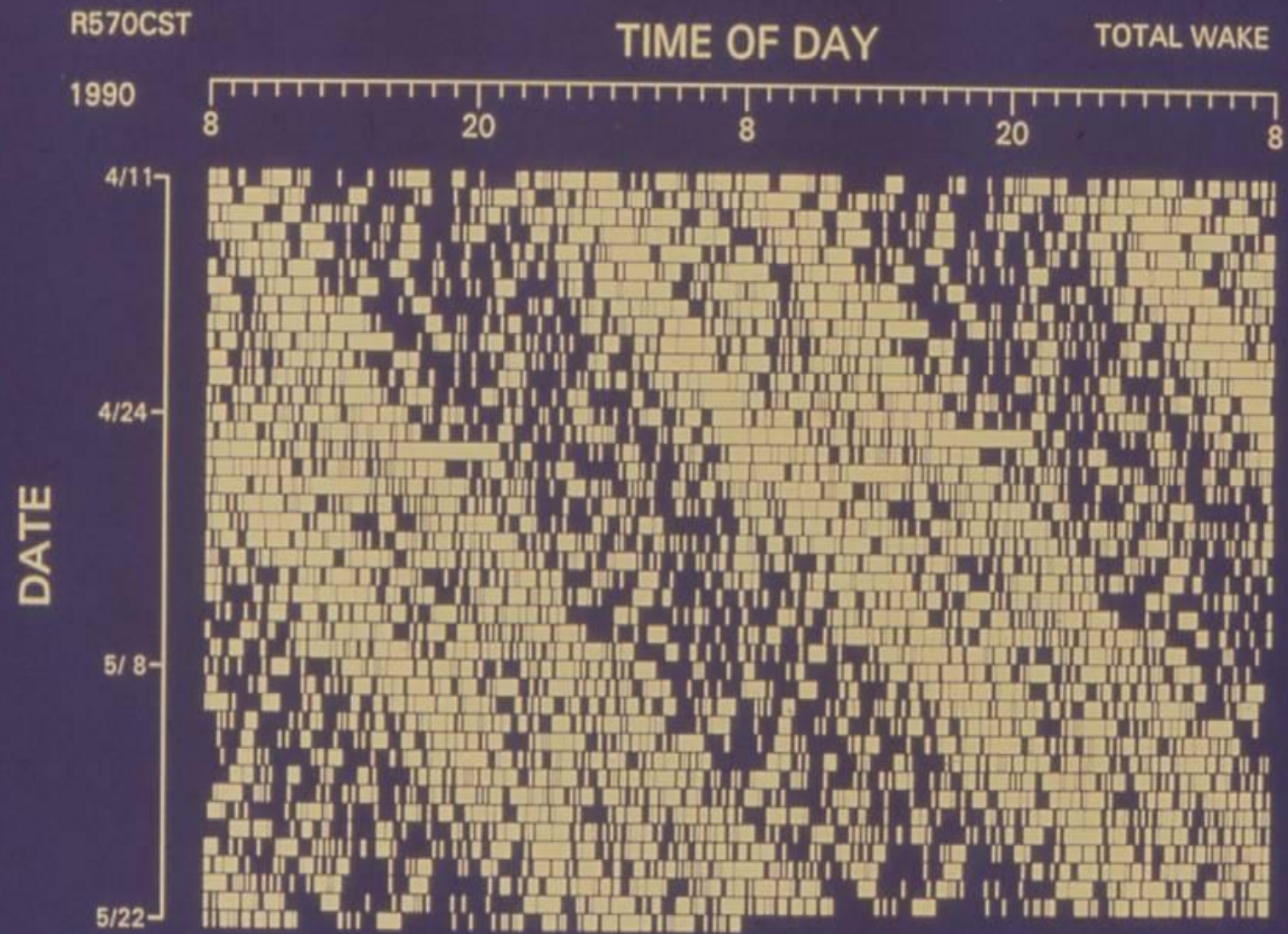


Courtesy of Dale Edgar PhD

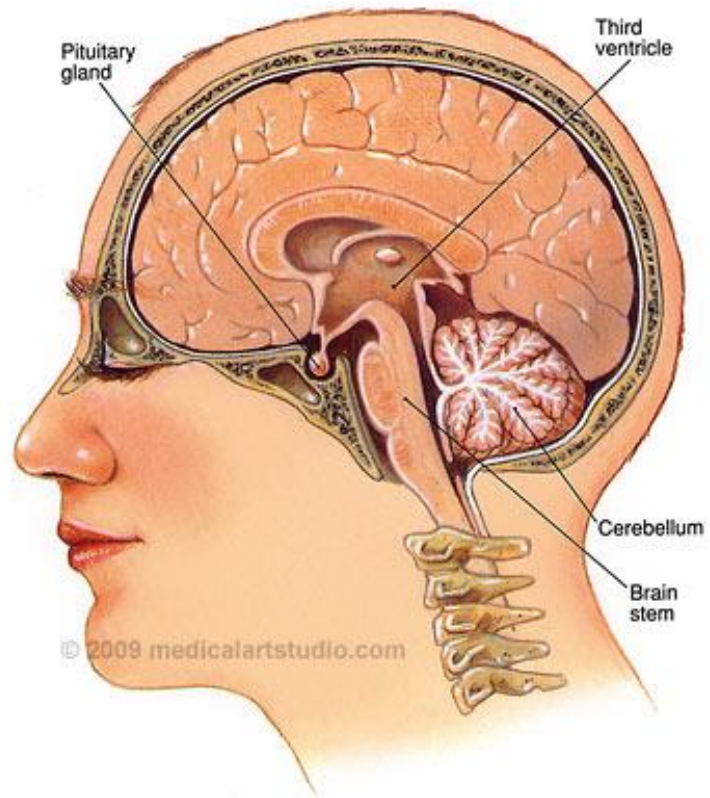


**2017 Nobel Prize in
Physiology/ Medicine
awarded for discoveries of
molecular mechanisms
controlling the circadian
rhythm**



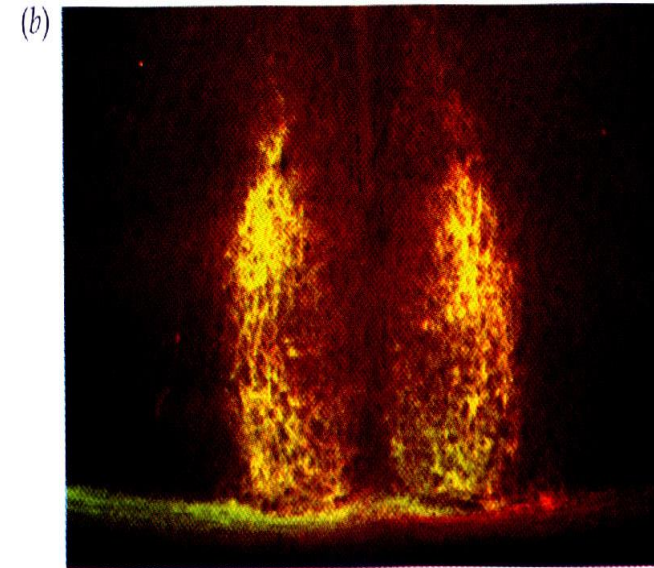
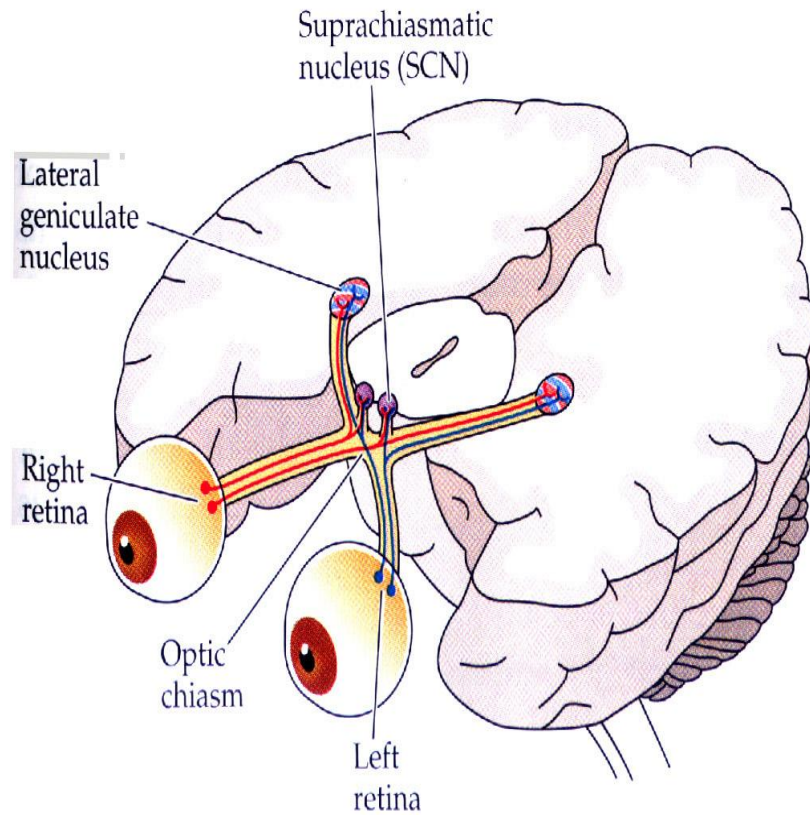


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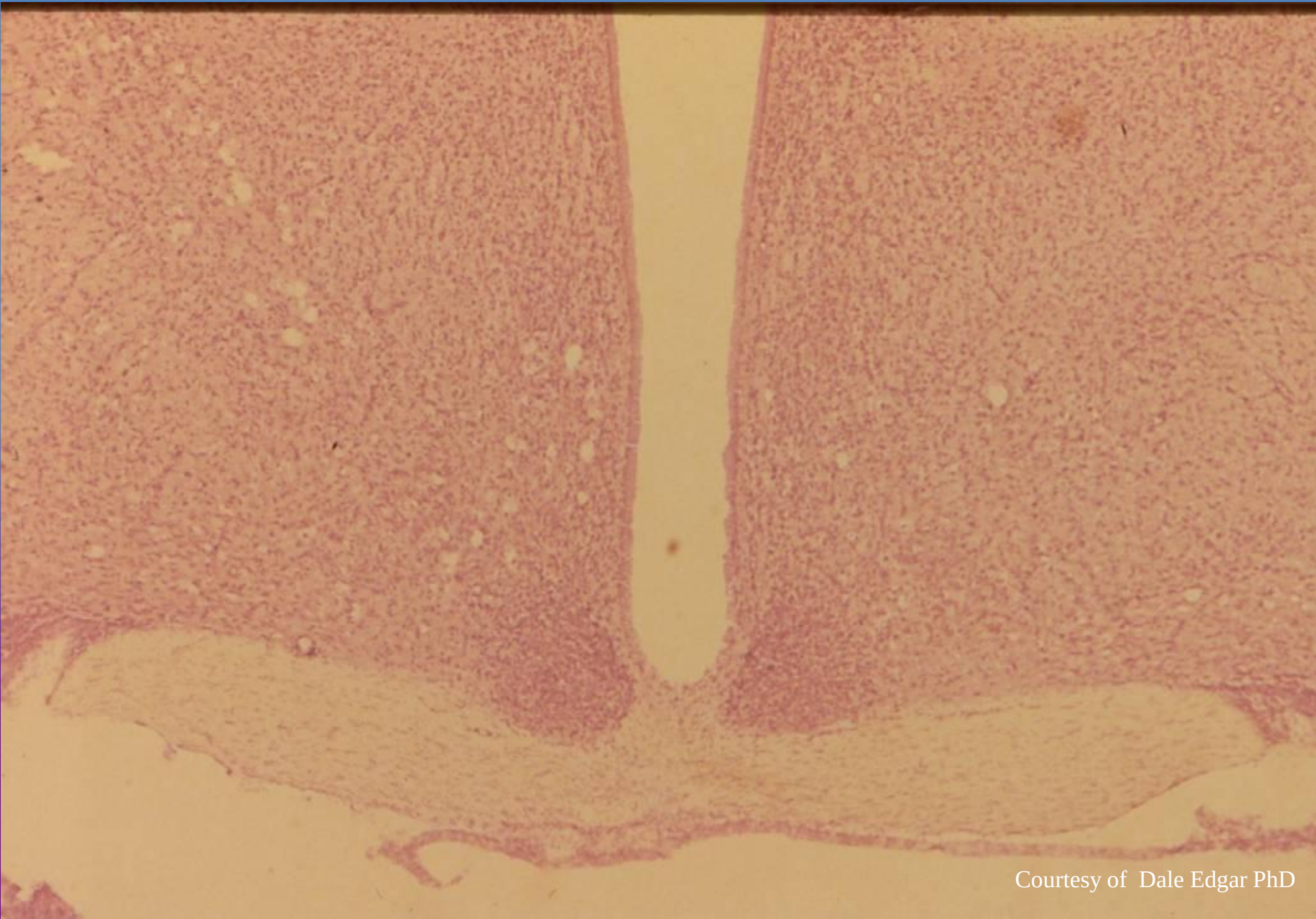


Sagittal Section

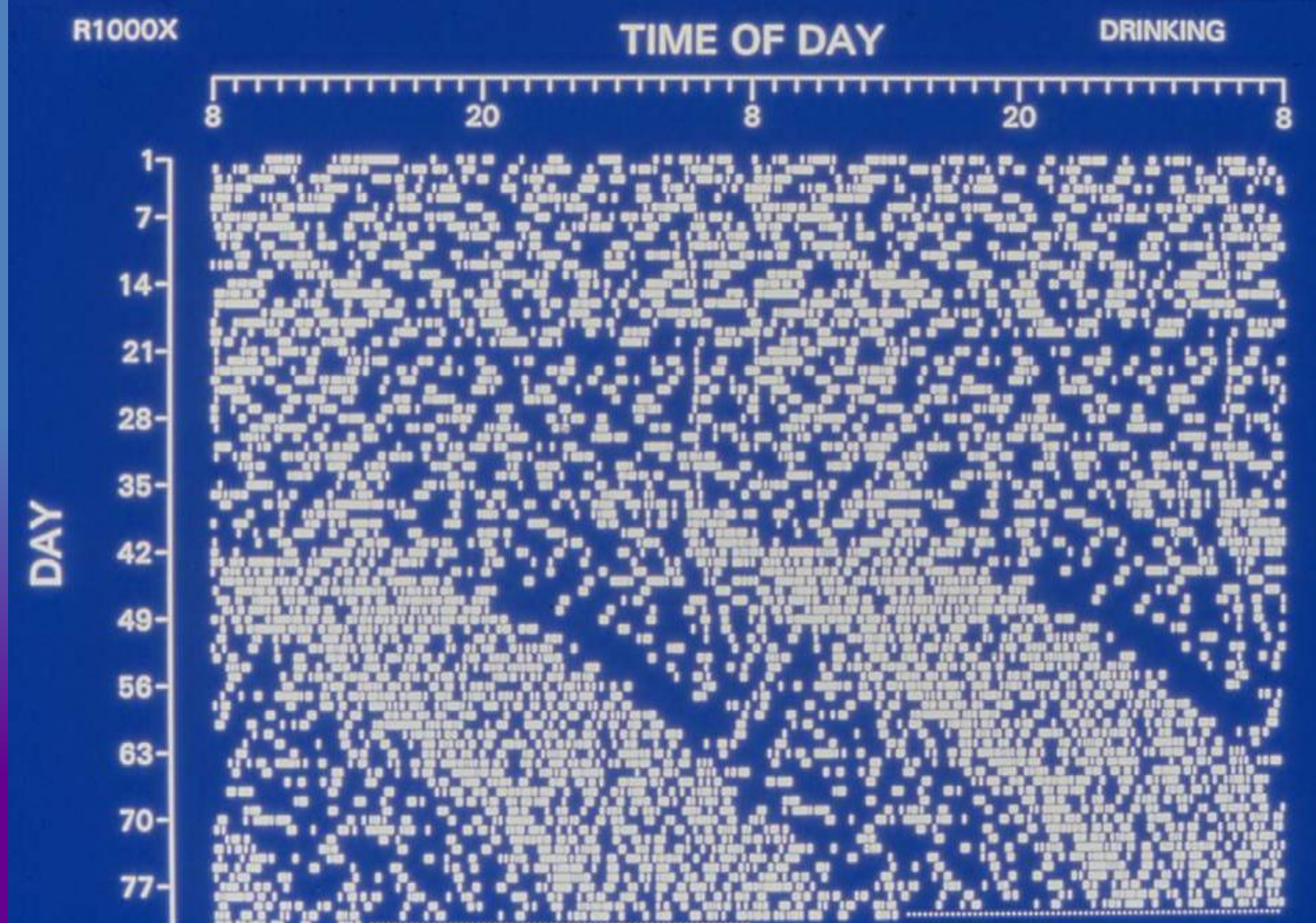


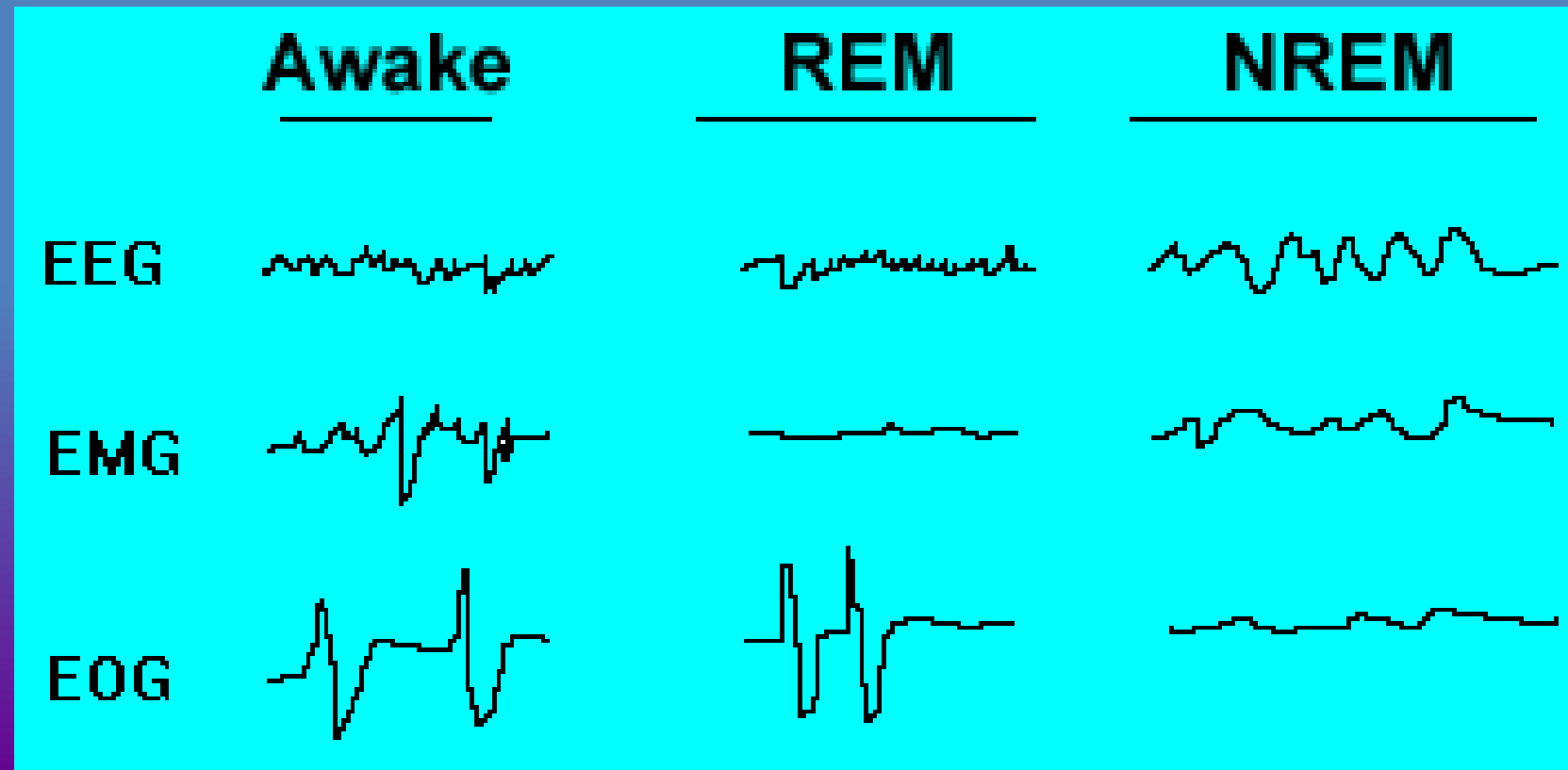


14.5 The Retinohypothalamic Pathway in Mammals (a) This pathway carries information about the light-dark cycle in the environment to the SCN. For clarity of synaptic connections, the SCN is shown proportionately larger than other features. (b) Axons (seen at the bottom of the image) from the left eye are labeled red, while those from the right are green. Both eyes project so diffusely to the two overlying SCN that they are outlined in yellow. (Photograph courtesy of Cynthia L. Jordan.)

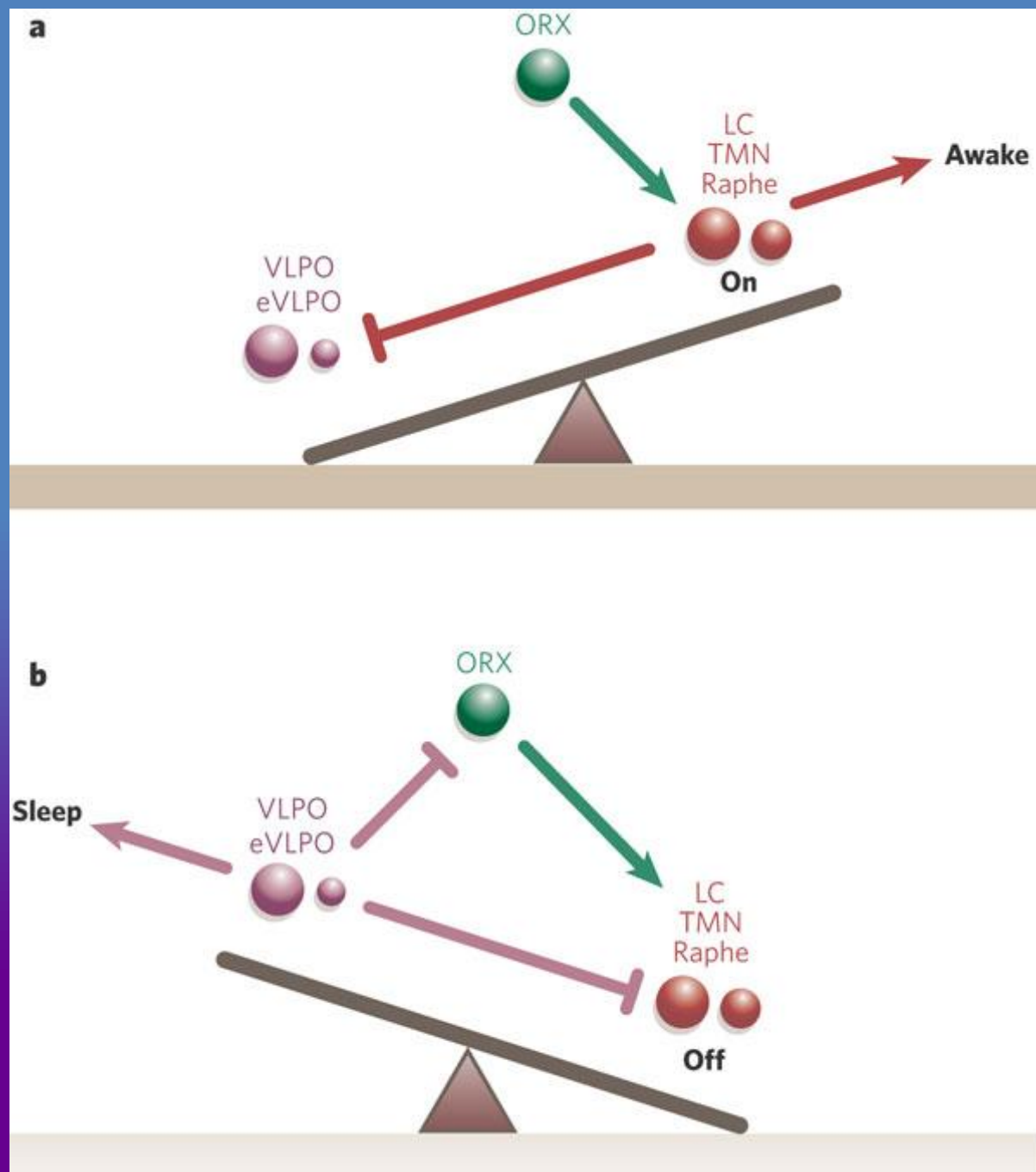


Courtesy of Dale Edgar PhD

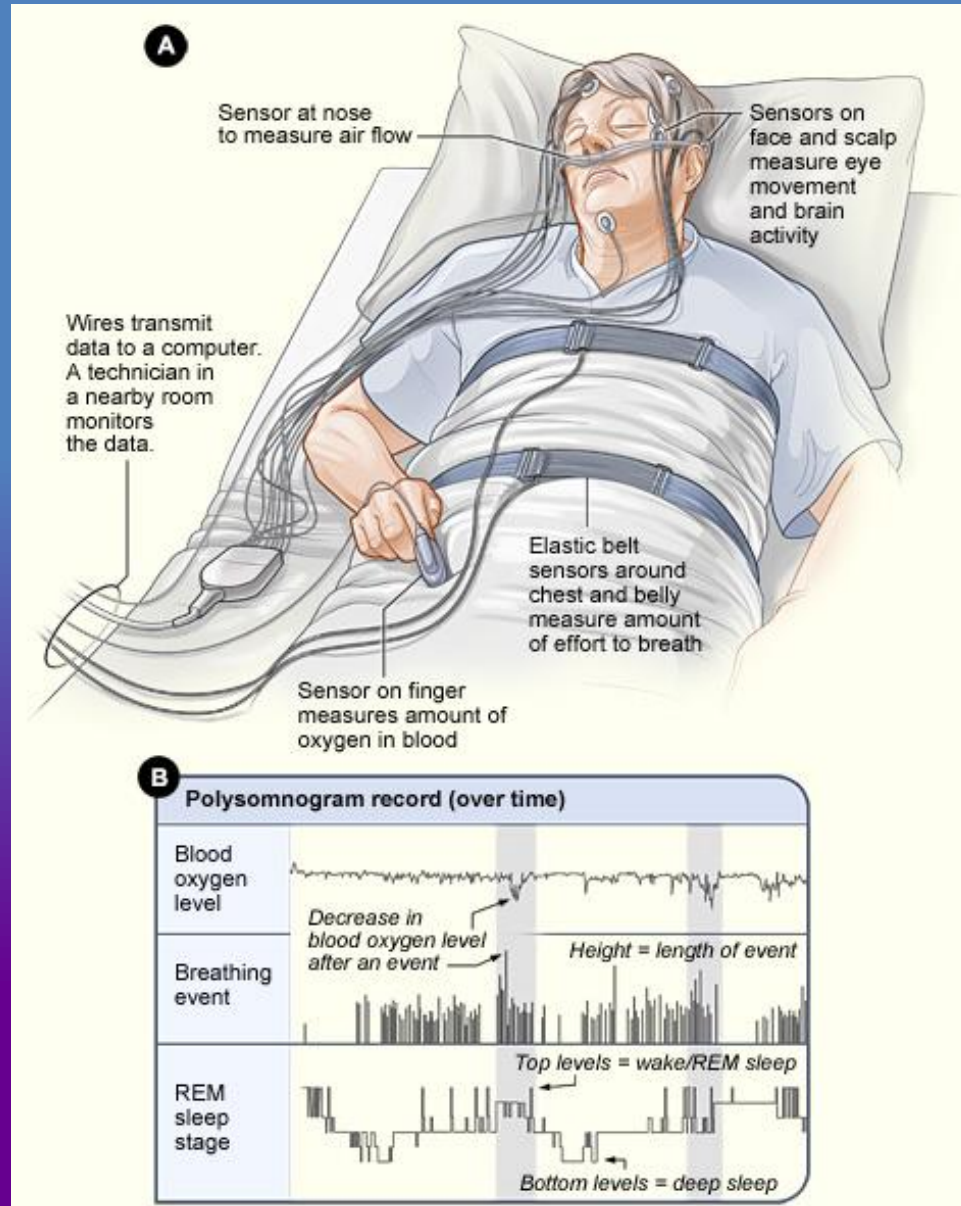




Sleep Flip Flop Switch



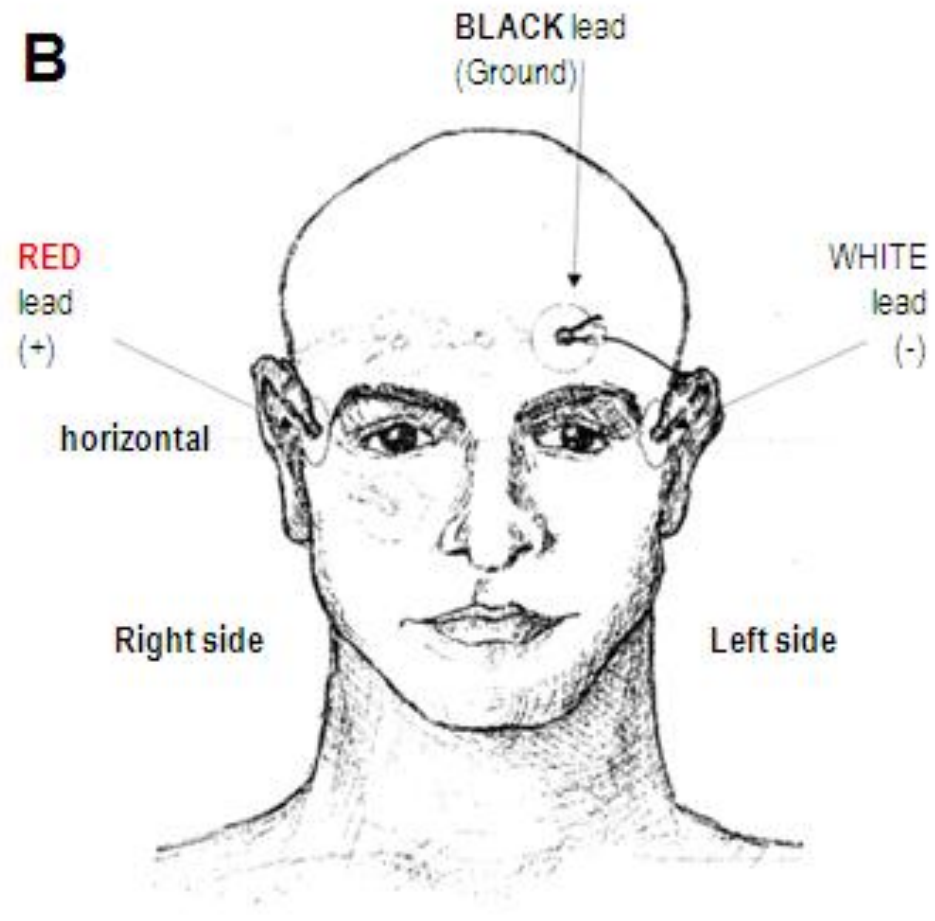
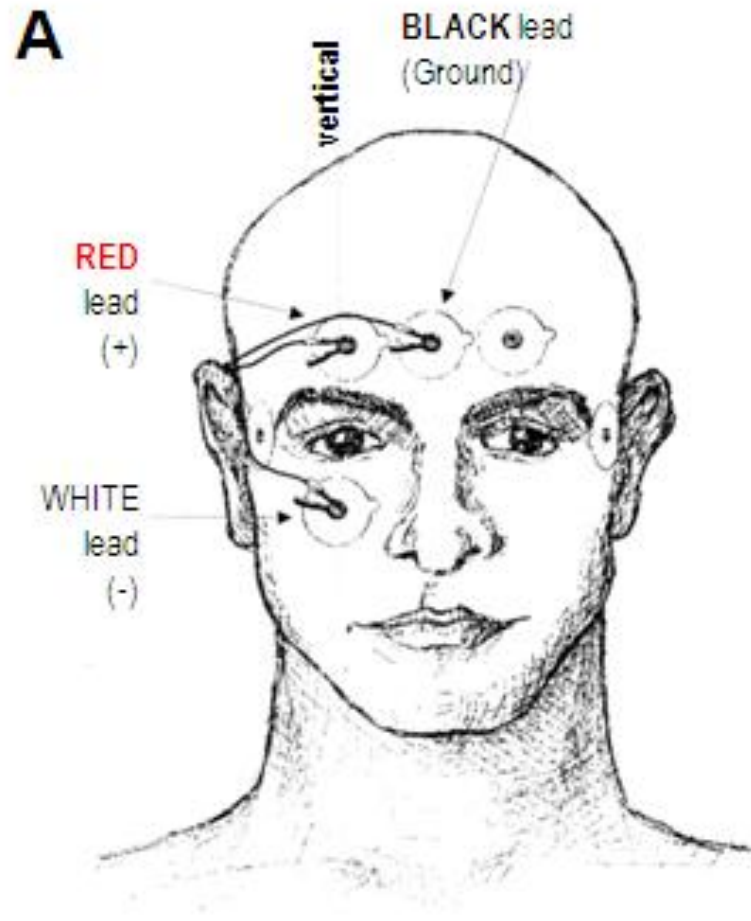
Polysomnography



How to measure eye movements



Electro Ocular Gram



EEG

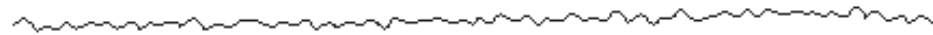
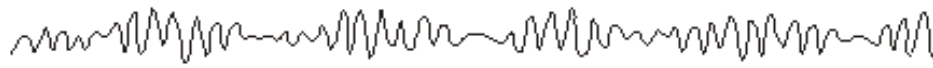
Beta (β) 13-30 Hz

Parietally and
frontally



Alpha (α) 8-13 Hz

Occipitally



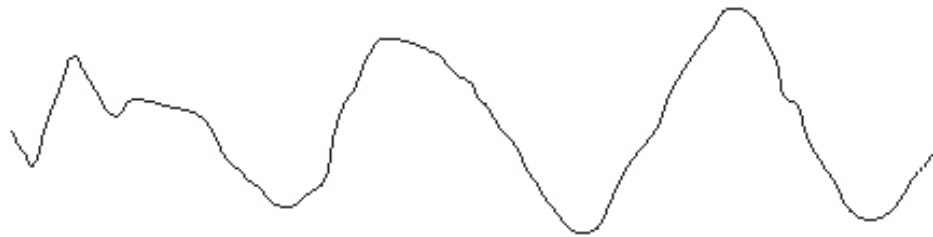
Theta (θ) 4-8 Hz

Children,
sleeping adults



Delta (δ) 0.5-4 Hz

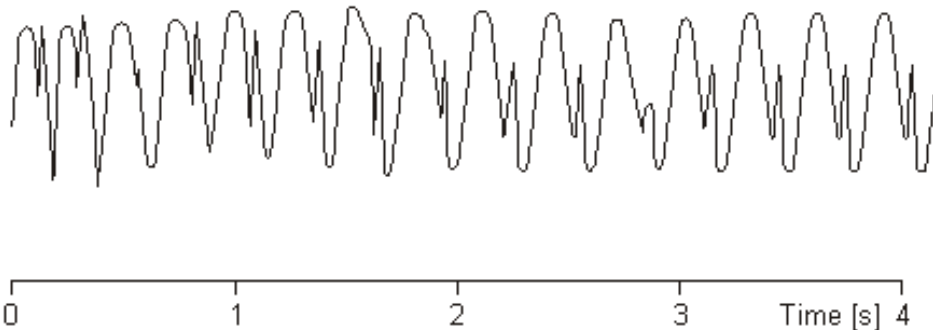
Infants,
sleeping adults



Spikes 3 Hz

Epilepsy -
petit mal

200
V [μ V]
100
0



Relaxed wakefulness

Alpha waves



Stage N1



Theta waves

Stage N2

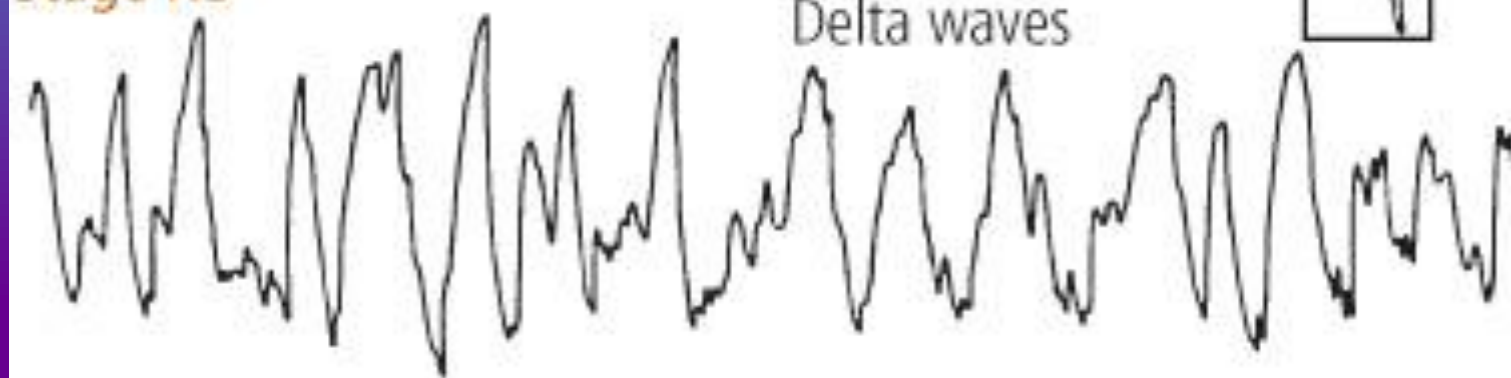


Sleep spindles

K-complex

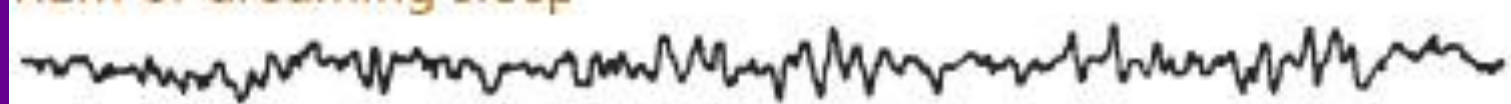


Stage N3

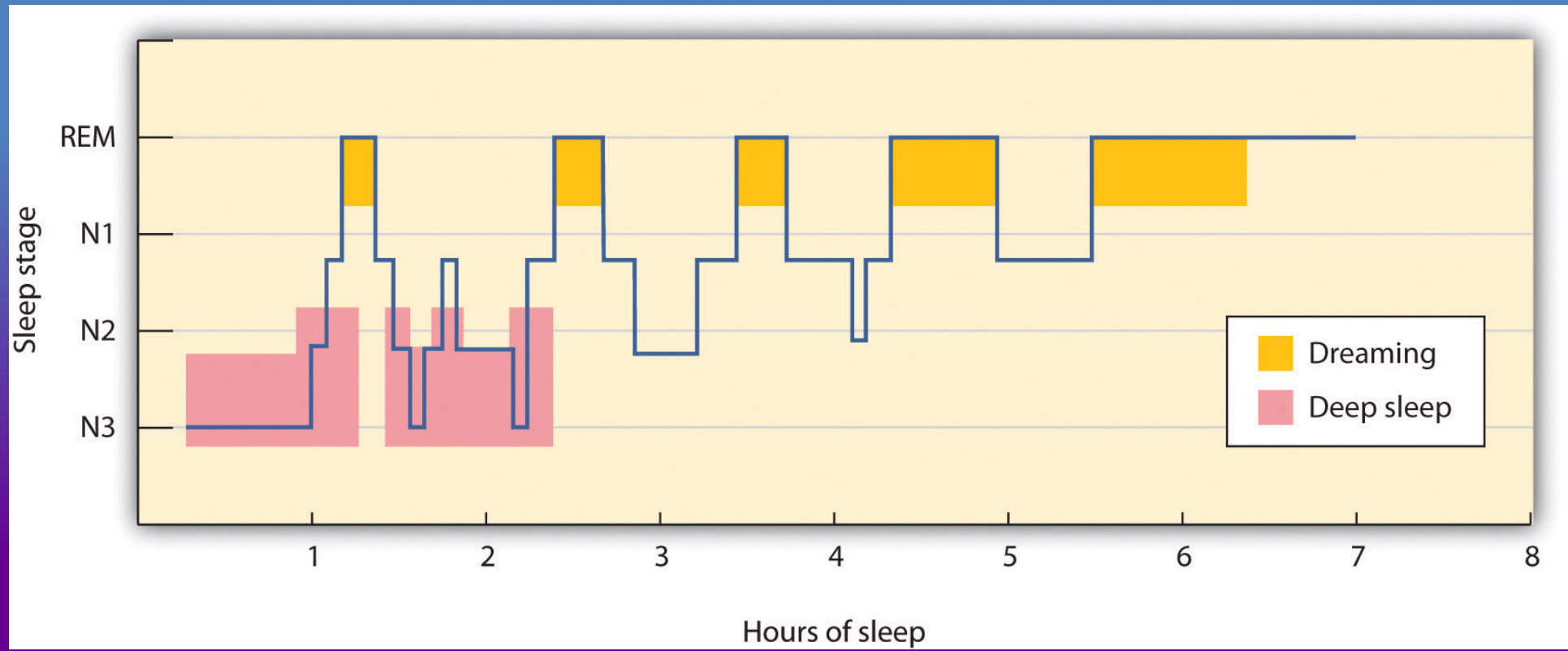


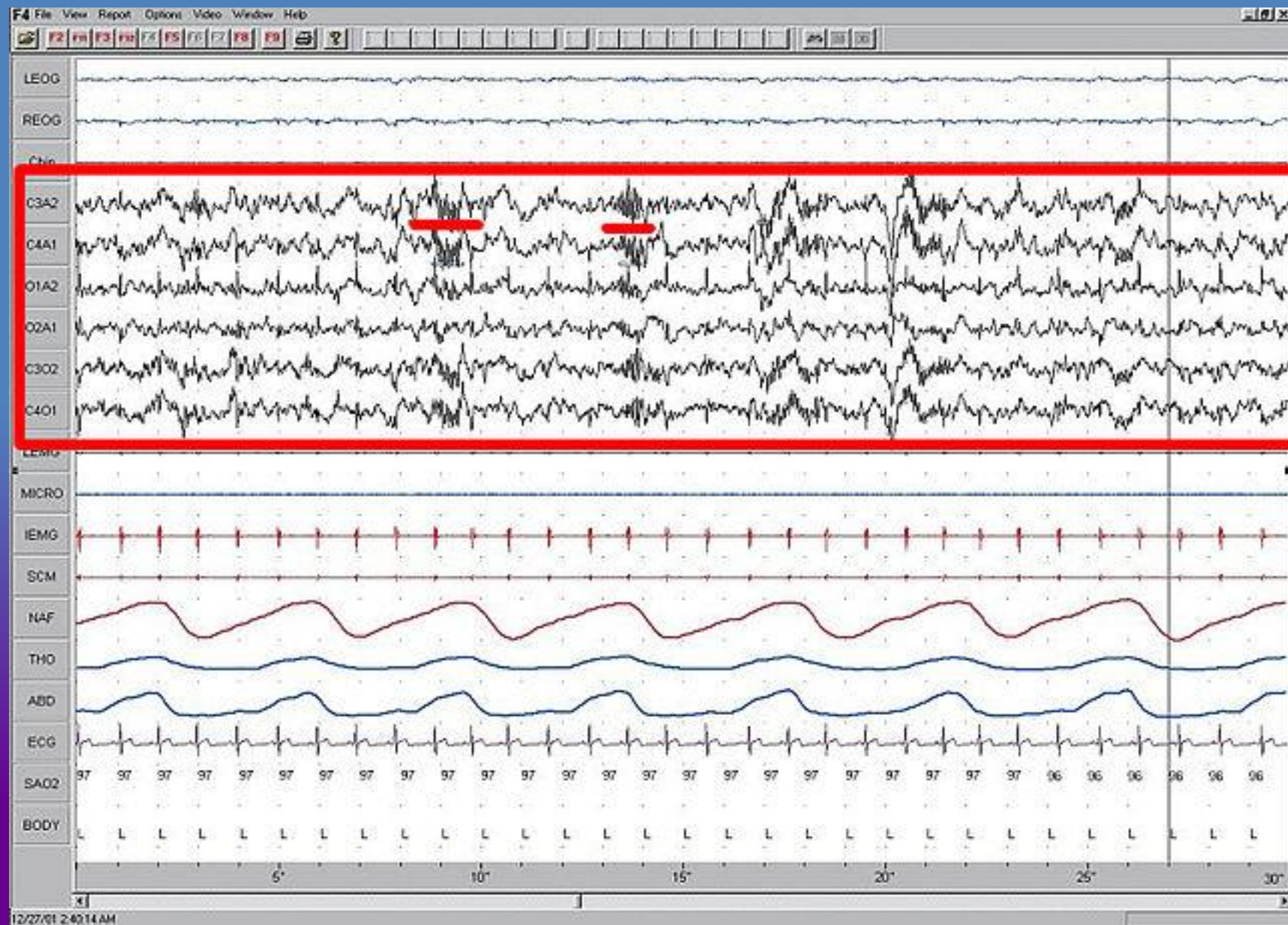
Delta waves

REM or dreaming sleep

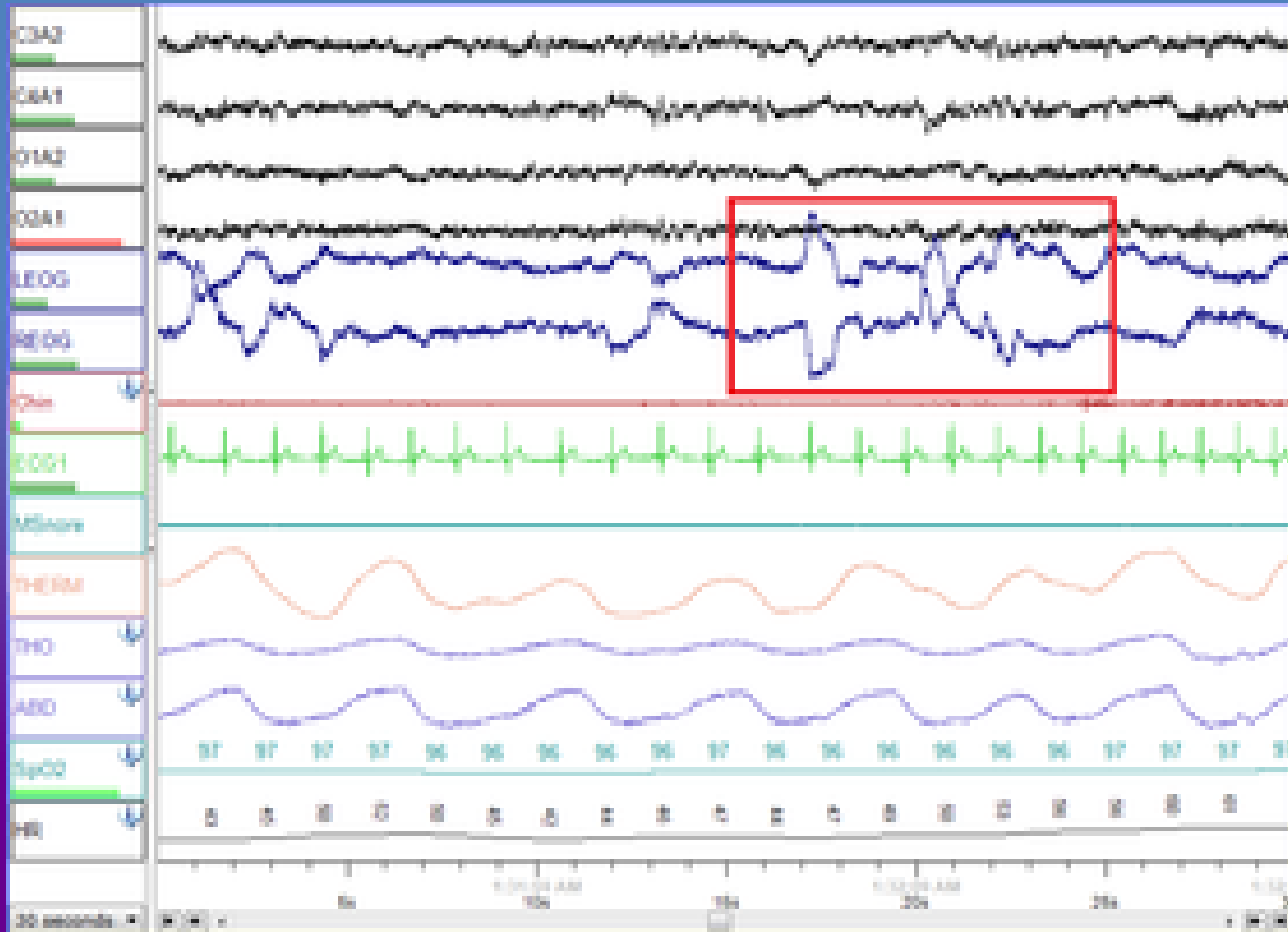


Sleep Architecture

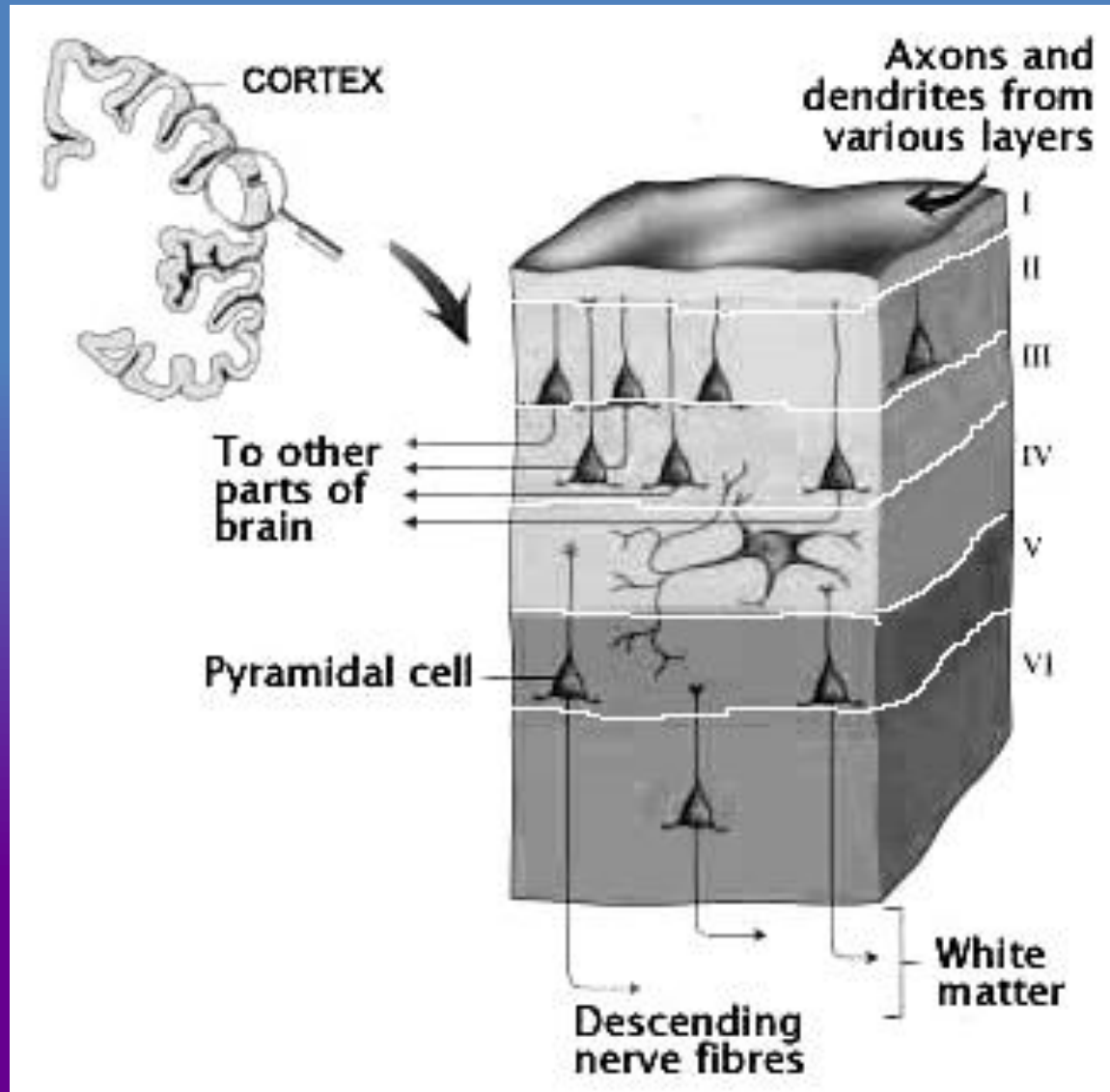




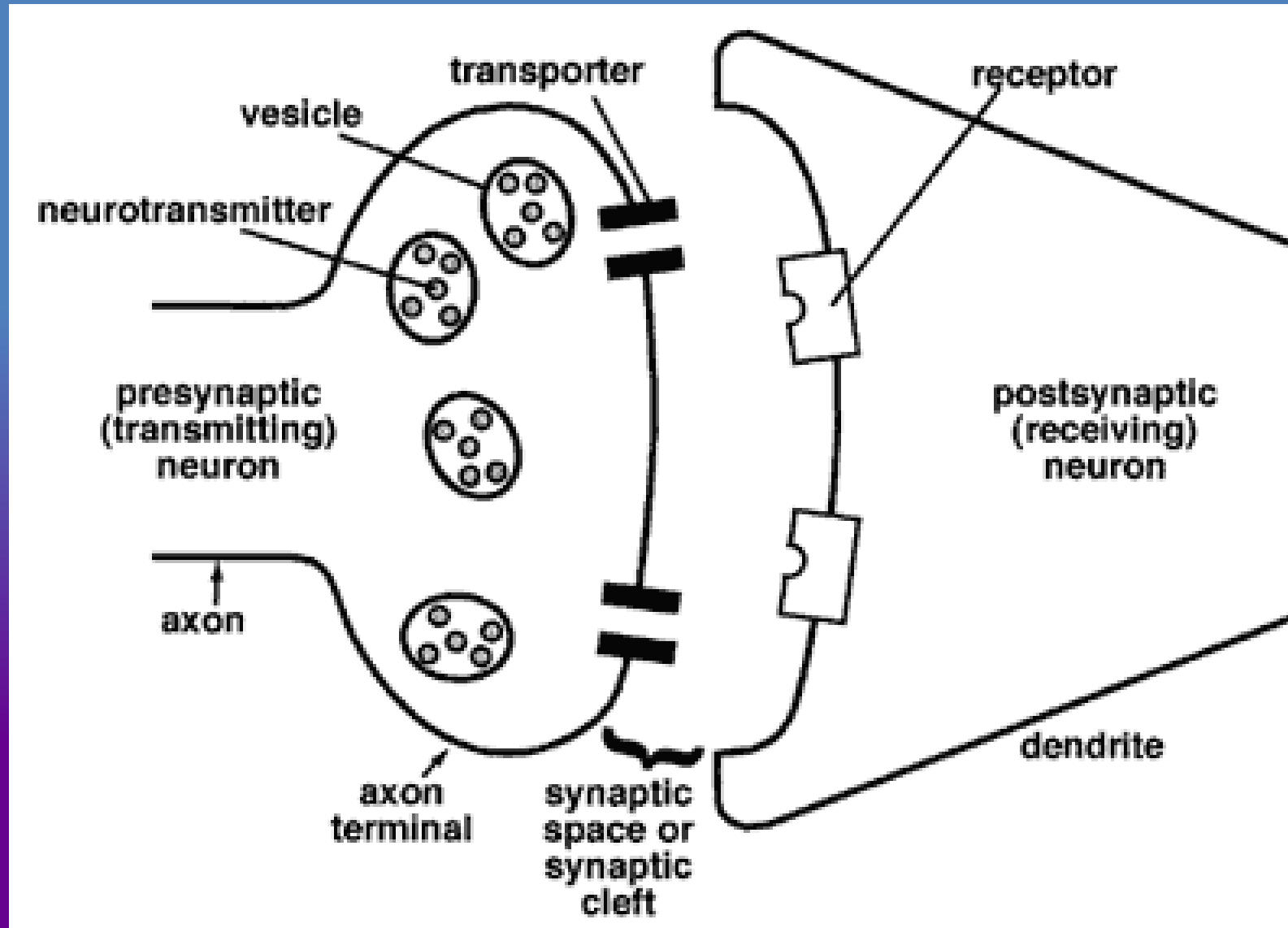
REM Sleep

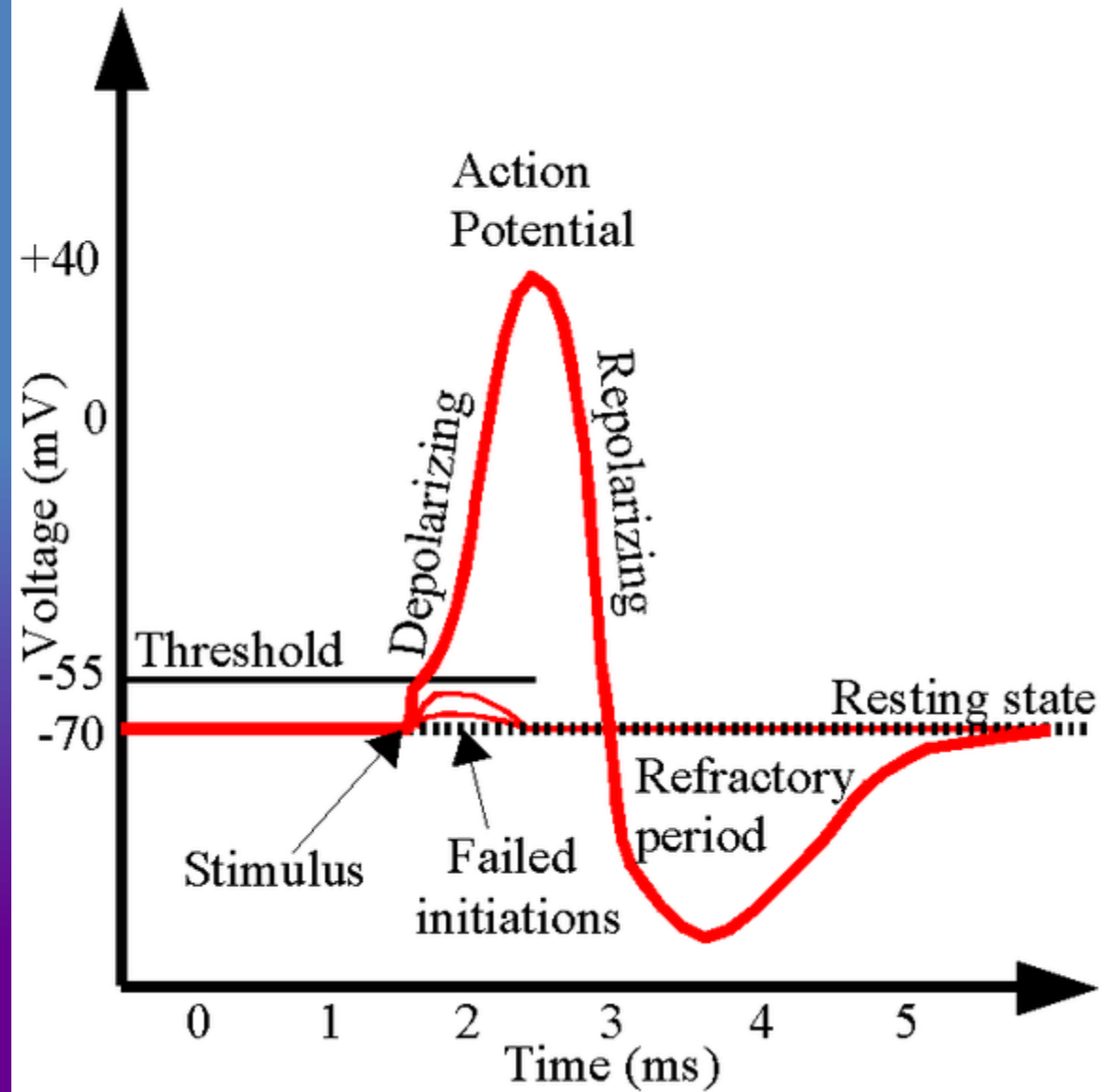


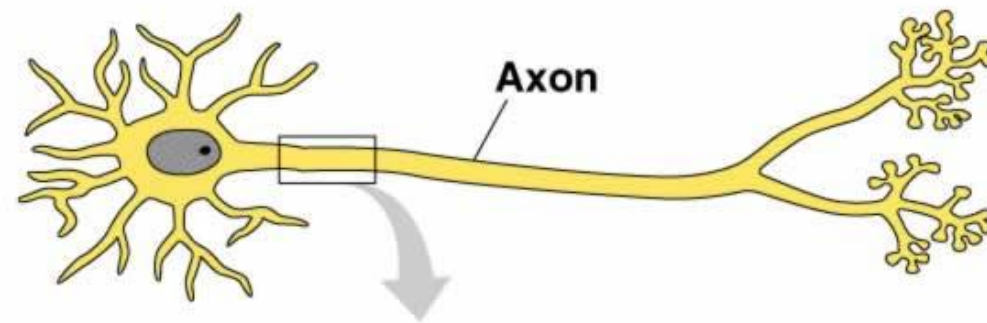
Neocortical Layers



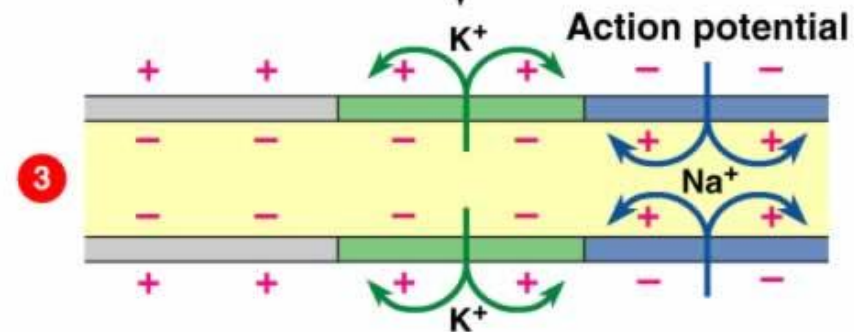
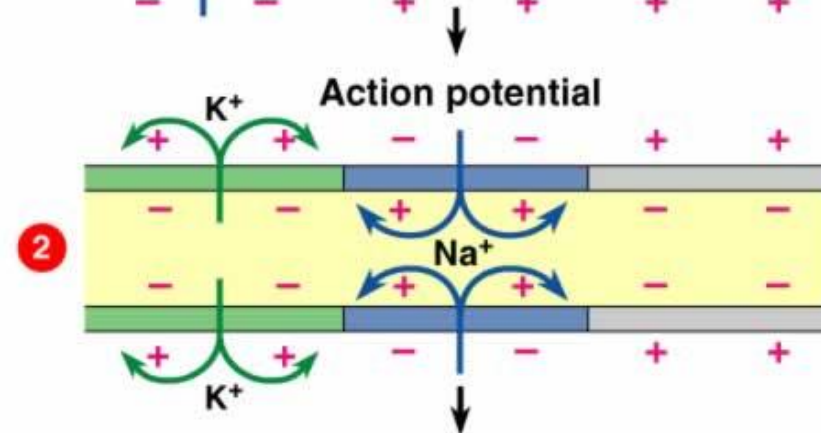
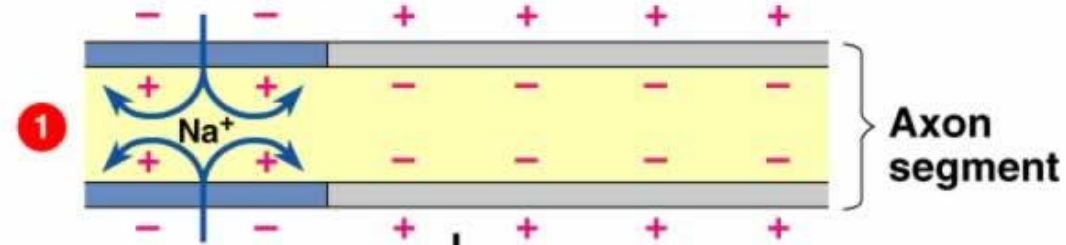
Synaptic Cleft

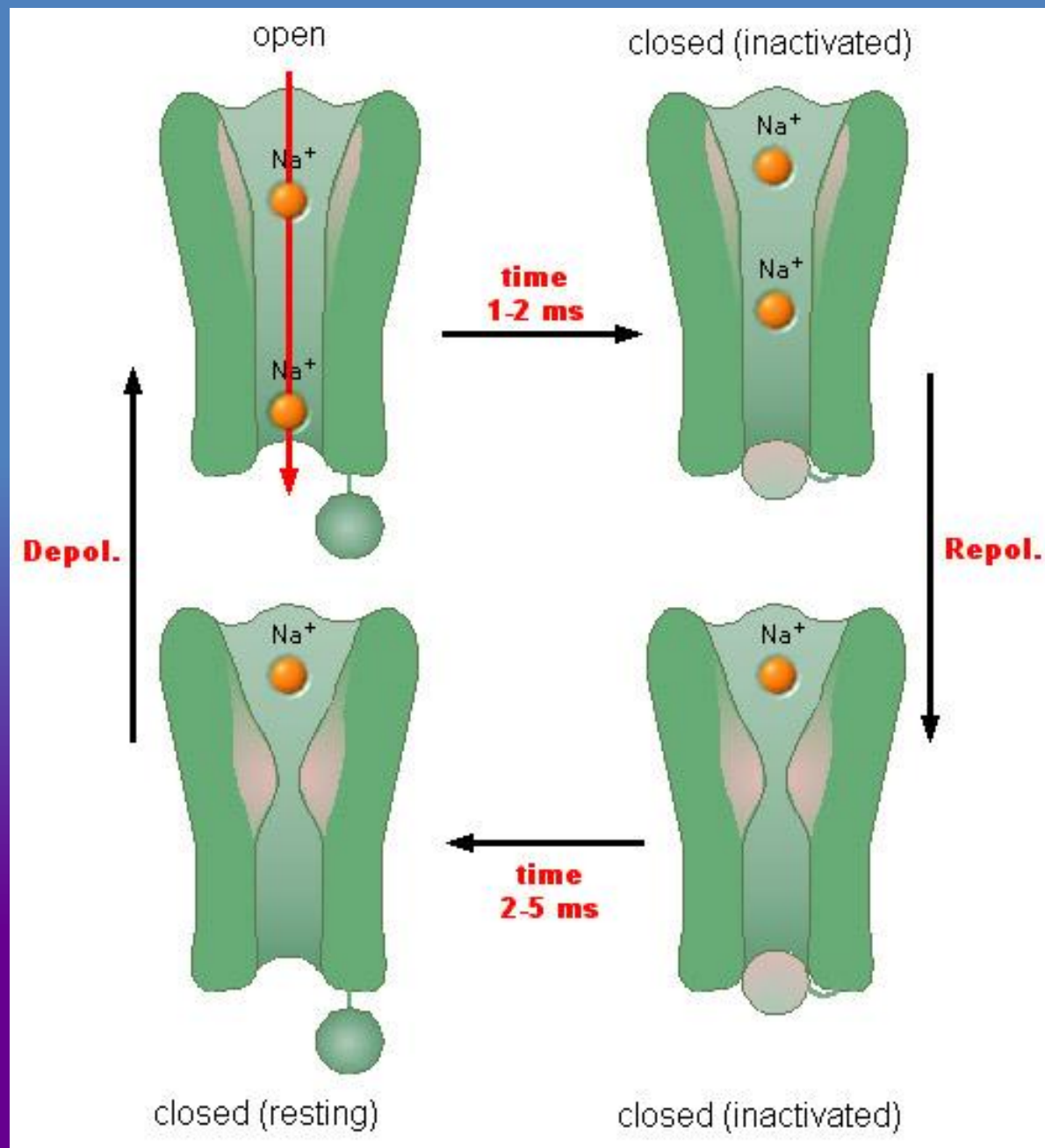




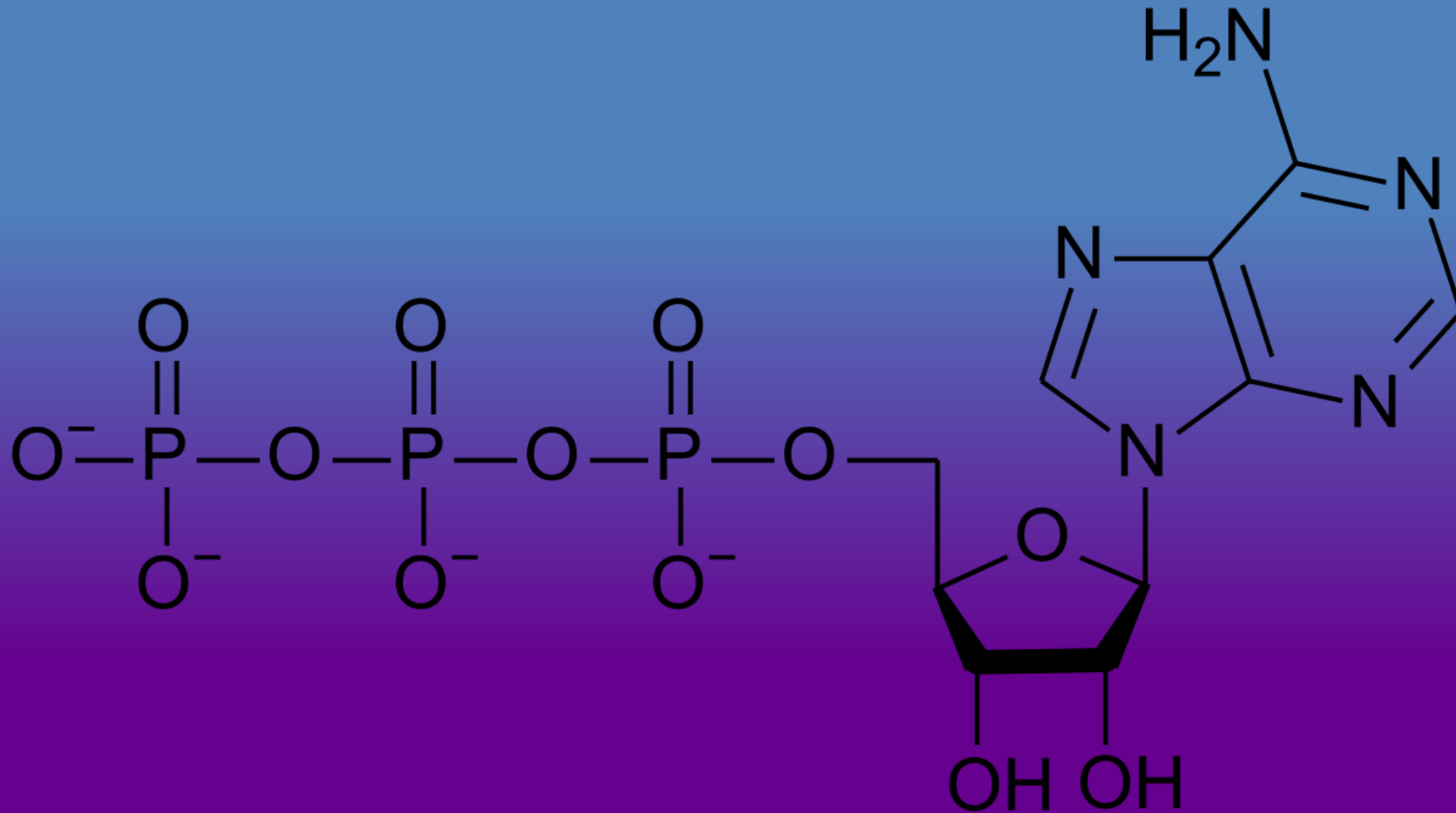


Action potential





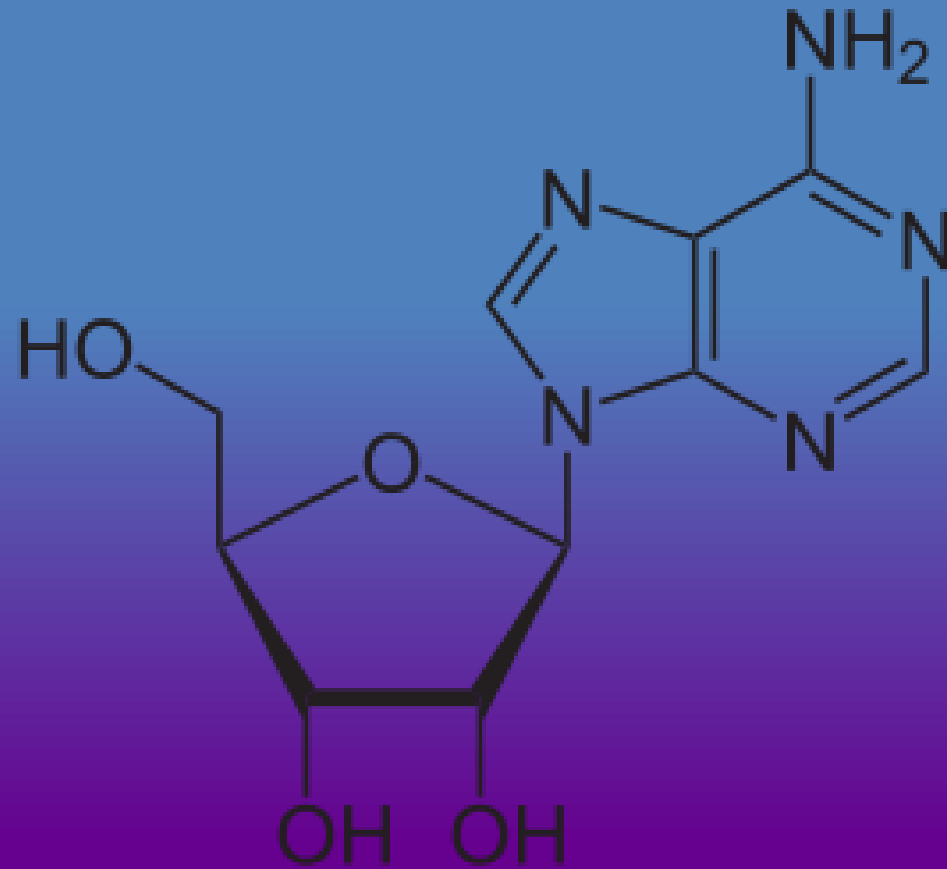
Adenosine Triphosphate ATP



Synaptic Homeostasis

- “sleep is for an energy surge”, a surge that permits energy-consuming anabolic processes, such as protein and fatty acid synthesis, to occur.
- Higher energy utilization during wake and its reversal during sleep, since increased synaptic potentiation during wake would contribute to high energy consumption and the surge in ATP during sleep would reflect the energy conserving processes such as synaptic downscaling during sleep.
- The availability of ATP in sleep could promote the changes related to synaptic plasticity and memory consolidation and remodeling during sleep

Adenosine



Marketing of 300mg Caffeine to Children



A Teenager Died After Drinking Too Much Caffeine

Drank a diet Mountain Dew, a cafe latte from McDonald's, and an energy drink within two hours, a coroner ruled
BuzzFeed May 16, 2017.



54 mg in can



77 mg per shot



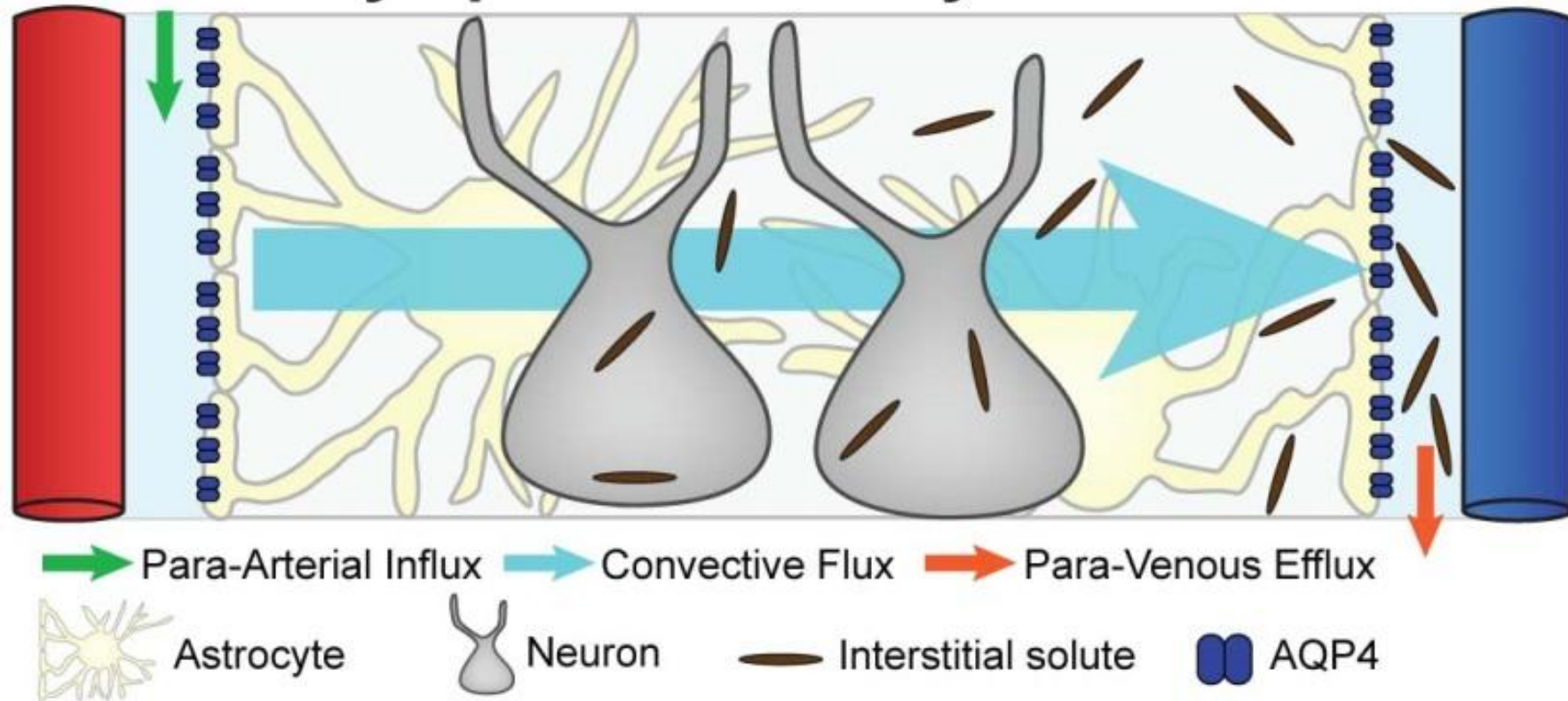
160 mg per 16 oz can

Glymphatics



Does sleeping wash our brains?

Glymphatic Pathway Function

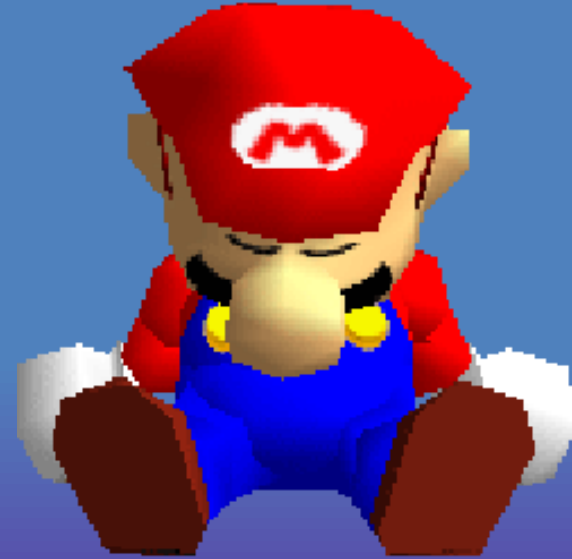


Can sleepiness be measured?



Behavioral signs of sleepiness seem obvious

- Yawning
- Ptosis
- Reduced behavioral activity
- Lapses in attention
- Closing eyes
- Head nodding



Factors such as motivation and stimulation can reduce the behavioral signs of sleepiness. This is why people may underestimate physiological sleepiness

Epworth Sleepiness Scale

Use the following scale to choose the **most appropriate number** for each situation:

- 0 = would **never** doze
- 1 = **slight chance** of dozing
- 2 = **moderate chance** of dozing
- 3 = **high chance** of dozing

It is important that you answer each question as best you can.

Situation	Chance of Dozing (0-3)
Sitting and reading _____	_____
Watching TV _____	_____
Sitting, inactive in a public place (e.g. a theatre or a meeting) _____	_____
As a passenger in a car for an hour without a break _____	_____
Lying down to rest in the afternoon when circumstances permit _____	_____
Sitting and talking to someone _____	_____
Sitting quietly after a lunch without alcohol _____	_____
In a car, while stopped for a few minutes in the traffic _____	_____
TOTAL	_____

Multiple Sleep Latency Test

- MSLT invented at Stanford and now used worldwide!
- Standardized test to measure sleepiness
- Consists of five 20- nap opportunities during the time expected to be awake
- Person is in bed and asked to go to sleep
- If they fall asleep within 20 minutes the nap is extended for 15 minutes to see if REM sleep occurs (SOREM)
- Average time to fall asleep is reported as the sleep latency, max is 20 minutes and minimum is zero.
- Scores of 8 minutes or less consistent with EDS



Questions?





Sleep is a paradox



Sleep Medicine ≠ Sleep Studies



Sleep Disorders

- Normal Sleep?
- Insomnia
- Obstructive Sleep Apnea
- Parasomnias
- Restless Leg Syndrome
- Narcolepsy
- Circadian Disorders i.e. Jet lag and Delayed Sleep Phase Syndrome



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Almost All U.S. Teens Are Sleep Deprived, Study Finds

HealthDay

Dec. 11, 2014 | 12:00 p.m. EST



+ More

Sleep History: 4 Elements

Amount

Quality

Timing

State of Mind

Tendencies are not destiny

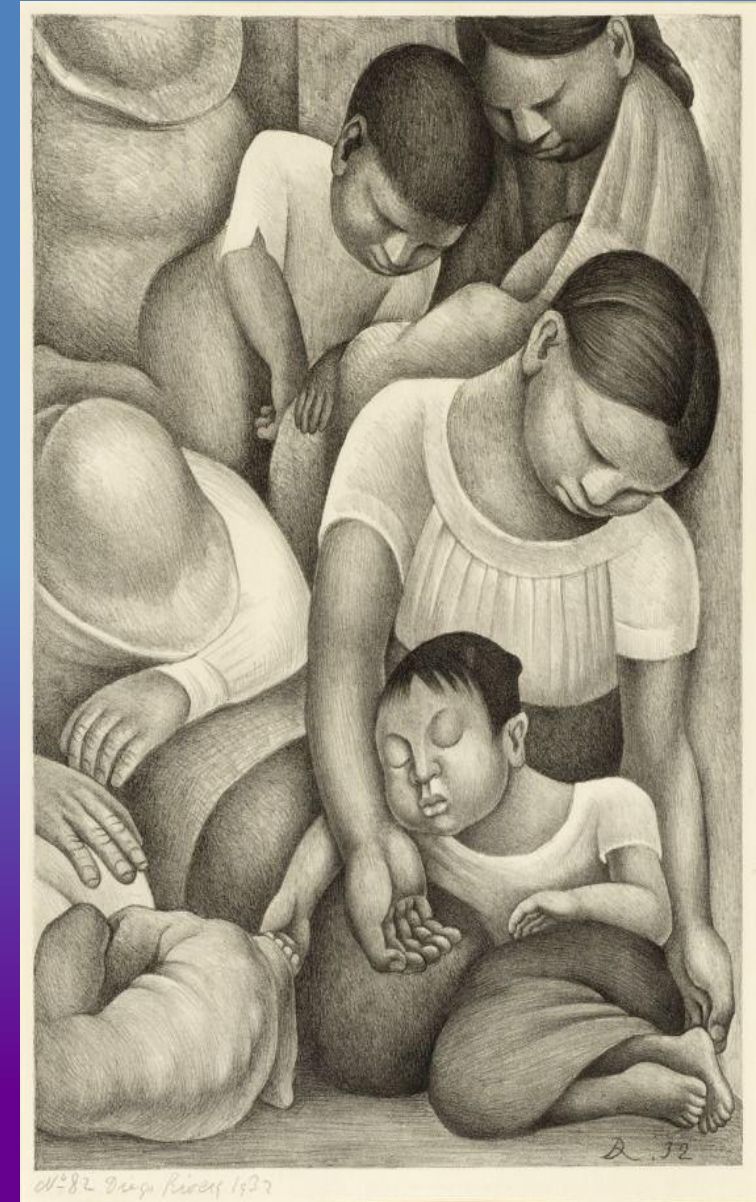
Grumpy Coffee Mug\$



Sleep is a paradox



Sleep Is The Most Natural And Powerful Form Of Self- care For All Of Us



Sueño by Diego Rivera

Functions of sleep

- Higher-order cognitive function involving mechanisms of learning and memory and synaptic plasticity



- Restorative function including maintenance of brain energy metabolism, macromolecular biosynthesis, and removal of metabolic waste

Flow ten times more active when the mice were asleep, as opposed to when they were awake

